

Weak Identification in Production Function Estimation: Diagnostics and Robust Inference*

Jorge de la Cal Medina
University of Amsterdam, Tinbergen Institute
`j.delacalmedina@uva.nl`

September 15, 2026

[CLICK HERE FOR NEWEST VERSION](#)

Abstract

Control-function estimators identify production elasticities from input variation that instruments predict and that the productivity control does not already absorb. We show that this residual variation can be arbitrarily small while point identification continues to hold, so that the elasticities are weakly identified in the sense of the GMM literature. The relevant object is a orthogonalized Jacobian, obtained by projecting the nonparametric productivity dynamics out of the second-stage Jacobian; its rank governs identification and the associated concentration matrix governs identification strength. Two structural channels drive it toward rank deficiency: because static inputs respond to productivity, quasi-differenced inputs are nearly collinear given lagged productivity, and proxy noise attenuates the identifying signal. Each channel has a limiting case in which identification fails outright, and near that boundary the continuously updated GMM estimator has a non-normal limiting distribution, so conventional Wald inference is invalid. We adapt identification-robust diagnostics, confidence sets, and specification tests to the control-function setting, and show in Chilean and U.S. data that identification strength varies enough across standard datasets to reverse conclusions about markups.

*I am grateful for helpful comments from Isaiah Andrews, Giovanni Angelini, Stephane Bonhomme, Filippo Biondi, Giuseppe Cavaliere, Aureo de Paula, Sabien Dobbelaere, Massimo Giuliadori, Giulio Gottardo, and Franc Klaassen, and from participants at the MInt Seminar at the University of Amsterdam, the Economics Seminar at the Vrije Universiteit Amsterdam, the Econometrics Seminar at the University of Bologna, the Econometrics Seminar at the University of Zurich, and the 2025 EARIE, EEA, IAAE, EWMES and IPDC conferences, and the 2026 IIOC (Boston). Special thanks to Haiqing Xu for detailed comments that substantially improved the identification analysis.

1 Introduction

Production functions are central objects in economics. Their estimation has a long history, yet the core econometric difficulty motivating early work remains central today: firms choose inputs in response to productivity shocks unobserved by the econometrician, rendering inputs endogenous. Control-function methods, developed by Olley and Pakes (1996) (OP), Levinsohn and Petrin (2003) (LP), and Akerberg et al. (2015) (ACF), address this endogeneity by using observable firm decisions—most commonly investment or intermediate inputs—to construct a proxy for latent productivity and deriving moment restrictions from firms’ optimization behavior. These procedures are now used extensively across empirical industrial organization, macroeconomics, labor, and international trade (Akerberg et al., 2007).

Point identification requires the moment conditions to uniquely pin down the structural parameters. Even when this condition is met, however, the moment conditions may contain only limited information about some parameters. Control-function estimators are naturally cast in a GMM framework (Wooldridge, 2009), and the GMM literature—building on Staiger and Stock (1997) and Stock et al. (2002)—has extensively studied what happens when identification is *weak*: the GMM criterion is nearly flat in economically relevant directions, the sampling distribution of the estimator departs substantially from its asymptotic normal approximation, and conventional Wald-type confidence intervals have incorrect coverage (Stock and Wright, 2000; Stock et al., 2002). Formally, weak identification corresponds to the moment Jacobian being near-singular, and identification-robust inference procedures (Stock and Wright, 2000; Kleibergen, 2005) deliver valid inference regardless of identification strength.

That this distinction is not merely theoretical in the present setting can be seen by applying the same estimator to two standard datasets. In Chilean plant data for food manufacturing, a Cobb–Douglas production function is sharply determined: the materials elasticity is 0.948 with an identification-robust 95% confidence set of $[0.923, 0.968]$. Applied to U.S. Compustat data for the same industry, the same procedure returns a labor elasticity of -0.992 and a capital elasticity of 1.696 , with the robust set for labor running from -4.79 to 1.81 . Neither application rejects its overidentifying restrictions, so by the diagnostics ordinarily reported in this literature both pass. What separates them is how much identifying variation survives the productivity control: after partialling out the remaining quasi-differenced inputs, the instruments predict the Chilean labor and materials quasi-differences with conditional first-stage F statistics of 481 and 529, against 0.015

and 0.091 in Compustat. The Compustat parameters are formally point identified and the moment conditions appear valid; the moments simply carry almost no information about the parameters, and conventional inference does not report this.

In the control-function setting, two distinct sources drive near-singularity of the Jacobian. First, the proxy may carry a weak signal about productivity—a consequence of optimization frictions, heterogeneous input prices, supply-and-demand shocks, or construction from accounting aggregates. We use “weak proxy” throughout to mean this *signal-attenuation* failure. This differs from the Levinsohn and Petrin (2003) notion of a weak proxy as a failure of strict monotonicity of the material-demand function, which would prevent the proxy from recovering productivity altogether. The LP invertibility failure eliminates identification; signal attenuation degrades identification strength continuously while preserving formal identification under mild conditions, as shown in Section 3.5. Second, near-functional dependence between inputs: because static inputs are freely adjusted in response to productivity, their variation is nearly a deterministic function of dynamic inputs, limiting the independent variation available to separately identify input elasticities. This is the continuous analogue of the exact collinearity that ACF identify as a source of non-identification, and it has the same consequences for inference even when it falls short of exact failure.

This paper makes three contributions. First, we use the moment Jacobian to characterize identification and its strength in the control-function setting. Because the nonparametric productivity-dynamics function $g(\cdot)$ enters the moment conditions as a nuisance, the identification of the structural parameters (β, ρ) depends on variation that is not already captured by $g(\cdot)$. Following Robinson (1988), this is made precise through the *orthogonalized Jacobian* Π in (14), obtained by projecting out the nuisance directions from the full second-stage Jacobian after standardizing the moments by their sampling variability: (β, ρ) is locally identified if and only if Π has full column rank. Near-rank-deficiency of Π is the precise sense in which identification is weak, and both sources—limited proxy signal and near-functional dependence—operate by driving Π toward rank deficiency through distinct structural channels. A Cobb–Douglas model with an AR(1) law of motion makes the two explicit and shows that identification strength is the product of a structural magnitude, such as the variance of labor optimization noise, and a geometric factor measuring how much of the resulting variation survives projection on the nuisance directions. This establishes a precise analogy between weak proxy informativeness and the weak-instrument problem in linear IV. We then show that when Π is nearly rank-deficient the continuously updated GMM estimator (CUE) has a nonstan-

dard sampling distribution and conventional Wald statistics are size-distorted in ways that standard pre-testing cannot detect. To do so we extend the framework of Stock and Wright (2000) to allow the part of the Jacobian that is due to the strongly identified parameters to depend on the weakly identified parameters, as occurs naturally in the production function model because the law-of-motion parameter multiplies the quasi-differenced inputs; both blocks then inherit non-normal limits, so Wald inference is unreliable even for parameters that are themselves well identified.

Second, we develop an identification-robust inference toolkit that integrates directly into the standard OP/LP/ACF estimation workflow. We implement three complementary procedures: (i) a bootstrap normality diagnostic (Angelini et al., 2024) that tests whether the CUE estimator’s sampling distribution conforms to the asymptotic normal approximation required for conventional inference to be valid; (ii) identification-robust confidence sets obtained by inverting the S-statistic (Stock and Wright, 2000)—the nonlinear GMM analogue of the Anderson–Rubin statistic—which maintain correct coverage regardless of identification strength and apply to the full parameter vector as well as to subsets and linear transformations of interest, such as returns to scale; and (iii) a weak-identification-robust specification test (Kleibergen, 2005) that remains pivotal under weak identification, which is what allows limited proxy informativeness to be distinguished from genuine model misspecification—a distinction the standard Hansen J -test cannot make when identification is weak.

Third, we document the empirical relevance of weak identification and show how it propagates to economically important objects. In Monte Carlo experiments calibrated to the ACF firm-dynamics model, the proposed robust procedures maintain correct coverage as proxy informativeness deteriorates, while conventional Wald inference becomes increasingly unreliable and the bootstrap diagnostic responds to the same deterioration; a rank test, by contrast, continues to reject underidentification throughout, since detecting full rank is much weaker than establishing distance from rank deficiency. In applications to Chilean plant-level data (ENIA, ISIC 311) and U.S. firm-level data (Compustat, NAICS 31), identification strength varies systematically with data quality: ENIA’s directly measured intermediate consumption yields stable estimates across estimators, tight robust confidence sets, and no rejection of bootstrap normality, whereas Compustat, where all key inputs are constructed from accounting aggregates, yields sign reversals across estimators, robust sets that are wide or unbounded, and rejection of normality for every elasticity. Because the Hansen test rejects in neither sample, nothing in the Compustat results points to a detected violation of the

moment conditions, and we attribute the difficulty there to a lower signal-to-noise ratio across all inputs rather than to any specific proxy invertibility failure. Finally, since production-approach markups are ratios of elasticities to expenditure shares, an interval for an elasticity maps directly into intervals for markups. Propagating the robust sets through this map leaves Chilean markups informative, while in Compustat it yields bounds that are uninformative about both the level and the dispersion of markups, and a nearly flat implied density where the plug-in density looks precise. A tight plug-in markup distribution can therefore be an artifact of ignoring identification uncertainty.

Three literatures meet here. The first concerns identification in control-function models. Akerberg et al. (2015) show that functional dependence can destroy identification of flexible-input elasticities in the first-stage regression of OP and LP, and restore a route to identification by delaying flexible-input elasticities to second-stage moments. Gandhi et al. (2020) show that for gross-output specifications this second-stage step can itself fail, since current flexible inputs respond to the productivity innovation, and propose an approach built on first-order conditions. Doraszelski and Li (2025) establish that identification survives noisy proxies under an index-sufficiency condition. Each of these results concerns whether identification holds or fails. The region between the two—identification holds, but the surviving variation is small enough that conventional inference misleads—is what we analyze, and it is the region in which the ACF fix leaves much applied work.

The second is weak identification in GMM. Since Staiger and Stock (1997) and Stock et al. (2002), weak identification has been documented in the New Keynesian Phillips curve (Mavroeidis, 2005; Kleibergen and Mavroeidis, 2009b), monetary policy rules, DSGE models (Andrews and Mikusheva, 2015; Canova and Sala, 2009), consumption-based asset pricing (Stock and Wright, 2000), and Euler equations (Yogo, 2004). We add control-function production functions to that list, identify the structural features that generate the problem, and show which robust tools transfer. The third is the reliability of production-approach markups. Raval (2023) documents systematic discrepancies between labor- and materials-based markups across datasets, and De Ridder et al. (2022) show that with financial-statement data revenue-based estimates need not identify markup levels, while arguing that dispersion and trends remain informative. Our concern is different in kind: weak identification of the underlying elasticities propagates to markups whatever the data source, and it affects precisely the dispersion and trend objects that De Ridder et al. (2022) treat as more reliable.

The paper proceeds as follows. Section 2 introduces the model and its GMM representation. Section 3 characterizes identification via the orthogonalized Jacobian, catalogues the ways it loses

rank, shows how each failure becomes a matter of degree, and derives the resulting asymptotics. Section 4 develops the inference toolkit. Section 5 reports the Monte Carlo experiments. Section 6 presents the empirical applications and the markup analysis. Section 7 concludes.

2 Model

We follow the canonical OP/LP/ACF model and cast it in a GMM framework to analyze identification formally.

For firm i in period t , log output satisfies

$$y_{it} = f(k_{it}, v_{it}) + \omega_{it} + \varepsilon_{it}, \quad (1)$$

where k_{it} denotes the d_k predetermined inputs chosen at $t-1$ (e.g. capital), v_{it} denotes d_v static inputs chosen at t (e.g. labor), ω_{it} is the (scalar) firm-specific productivity observed by the firm but not by the econometrician, and ε_{it} is an idiosyncratic shock realized after input choices. This nests Cobb–Douglas, translog, and CES.

Productivity follows a first-order Markov process, such that its conditional distribution depends only on the previous realization.

Assumption 1 (First Order Markov).

$$P(\omega_{it} | \omega_{it-1}, \omega_{it-2}, \dots) = P(\omega_{it} | \omega_{it-1}). \quad (2)$$

This implies that the law of motion can be decomposed into its conditional expectation and an unpredictable innovation:

$$\omega_{it} = g(\omega_{it-1}) + \xi_{it} \quad (3)$$

where $g(\omega_{it}) = \mathbb{E}[\omega_{it} | \omega_{it-1}]$ and the innovation satisfies $\mathbb{E}[\xi_{it} | \omega_{it-1}] = 0$.

Let z_{it} be firm i 's choice vector at t , which includes k_{it+1} and v_{it} . Firms observe ω_{it} when choosing z_{it} ; ε_{it} is realized after these choices and ξ_{it} between $t-1$ and t .

Assumption 2 (Timing and sequential exogeneity). For each firm i , period t , and $\tau \leq t$:

$$\mathbb{E}[\varepsilon_{it} \mid z_{i\tau}] = 0, \quad (4)$$

$$\mathbb{E}[\xi_{it} \mid z_{i\tau-1}] = 0. \quad (5)$$

The key assumption of the control function approach is that observable inputs chosen by the firm serve as proxies for ω_{it} . Let $p_{it} \subset z_{i\tau}$, $\tau \leq t$, denote the proxy vector.

Assumption 3 (Proxy relationship). There exists a function h_t such that

$$\omega_{it} = h_t(p_{it}). \quad (6)$$

This is the scalar-unobservable condition standard in OP/LP/ACF: ω_{it} is fully determined by p_{it} .¹

Identification follows two stages. Assumptions 2 and 3 imply the first-stage regression

$$\mathbb{E}[y_{it} \mid p_{it}] = f(k_{it}, v_{it}) + h_t(p_{it}). \quad (7)$$

Substituting Assumption 3 into the production function gives $y_{it} = f(k_{it}, v_{it}) + h_t(p_{it}) + \varepsilon_{it}$. Taking expectations conditional on p_{it} and applying $\mathbb{E}[\varepsilon_{it} \mid p_{it}] = 0$ —which follows from Assumption 2 since $p_{it} \subset z_{i\tau}$ for $\tau \leq t$ —yields (7). It is usually imposed that $(k_{it}, v_{it}) \subset p_{it}$, hence f and h_t are not separately identified in this first stage; only the composite is recovered. The second stage uses the Markov law of motion for productivity together with sequential exogeneity to obtain

$$\mathbb{E}[y_{it} \mid z_{it-1}] = \mathbb{E}[f(k_{it}, v_{it}) + g(\mathbb{E}[y_{it-1} \mid p_{it-1}] - f(k_{it-1}, v_{it-1})) \mid z_{it-1}]. \quad (8)$$

Substituting the Markov law of motion (3) into the production function yields $y_{it} = f(k_{it}, v_{it}) + g(\omega_{it-1}) + \xi_{it} + \varepsilon_{it}$. Under Assumption 3, $\omega_{it-1} = h_{t-1}(p_{it-1})$ is z_{it-1} -measurable, and the first stage identifies $h_{t-1}(p_{it-1}) = \mathbb{E}[y_{it-1} \mid p_{it-1}] - f(k_{it-1}, v_{it-1})$. Taking expectations conditional on z_{it-1} and applying $\mathbb{E}[\xi_{it} \mid z_{it-1}] = \mathbb{E}[\varepsilon_{it} \mid z_{it-1}] = 0$ from Assumption 2 yields (8), which provides the basis for identifying f .

¹We discuss in Section 3.5 how this condition can be relaxed.

2.1 Parametric specification

We parametrize f , g , and h_t using polynomial sieves following Wooldridge (2009). The production function is

$$f(k_{it}, v_{it}) = \alpha_f + x'_{it}\beta,$$

with x_{it} collecting all monomials in (k_{it}, v_{it}) up to degree L_f ; $L_f = 1$ gives Cobb–Douglas and $L_f = 2$ gives translog. The dimension of β is then $d_\beta = \binom{L_f + d_k + d_v}{d_k + d_v} - 1$, since α_f captures the constant separately. The law of motion is

$$g(u) = \alpha_g + b(u)'\rho,$$

with $b(u) = (u, \dots, u^{L_g})'$, so that $\dim(\rho) = L_g$.

The control function is

$$h_t(p_{it}) = \alpha_h + r'_{it}\lambda,$$

where r_{it} collects all monomials in the components of p_{it} up to degree L_h . The dimension of λ is, if $p_{it} \in \mathbb{R}^{d_p}$, $d_\lambda = \binom{L_h + d_p}{d_p} - 1$, again excluding the constant.

Substituting these parametric forms into (7) and (8) yields the estimating equations from which we define the residuals. As mentioned above, the first-stage regression does not identify (β, λ) separately, since x_{it} is a subvector of r_{it} . Instead it identifies the composite parameter $\gamma \in \mathbb{R}^{d_\lambda}$, which satisfies $r'_{it}\gamma = x'_{it}\beta + r'_{it}\lambda$:

$$y_{it} = \alpha_1 + r'_{it}\gamma + u_{1,it}. \quad (9)$$

α_1 is a composite intercept, since the intercepts α_f, α_h are not identified separately. The second-stage equation is

$$y_{it} = \alpha_2 + x'_{it}\beta + b(r'_{it-1}\gamma - x'_{it-1}\beta)'\rho + u_{2,it}, \quad (10)$$

where $r'_{it-1}\gamma - x'_{it-1}\beta$ is the observable dynamic index (lagged predicted output net of the production component). Again, α_2 is a composite intercept.

Let $\theta := (\alpha_1, \alpha_2, \beta', \gamma', \rho')' \in \Theta \subset \mathbb{R}^{d_\theta}$ and use (9), (10) to define the stacked residual function $u_{it}(\theta) = (u_{1,it}(\theta), u_{2,it}(\theta))'$. With instrument vectors $w_{1,it} = (1, r'_{it})'$ and $w_{2,it} = (1, \tilde{r}'_{it-1})'$ stacked block-diagonally as W_{it} , the moment contribution is $\phi_{it}(\theta) = W'_{it}u_{it}(\theta)$, where $\tilde{r}_{it-1}$ collects all monomials in r_{it-1} up to degree d_ρ .

2.2 Special case: AR(1)

In the special case of a linear law of motion with $d_\rho = 1$ (AR(1) law of motion), the two equations collapse to the single estimating equation

$$y_{it} = \alpha_s + x'_{it}\beta + \rho(y_{it-1} - x'_{it-1}\beta) + u_{s,it}, \quad (11)$$

which obviates separating identification in two stages separately and reduces the parameter space considerably, by d_λ . We can use (11) to define residual function $u_{s,it}(\theta)$. With instruments $w_{s,it} = (1, r_{it-1})$, the moment contributions are then $\phi_{it}(\theta) = w_{s,it} u_{s,it}(\theta)$.

2.3 GMM Estimation

We estimate the model by GMM. Let $\phi_i(\theta) := \frac{1}{T-1} \sum_{t=2}^T \phi_{it}(\theta)$ aggregate moment contributions to the firm level, and let $\bar{\phi}_n(\theta) := \frac{1}{n} \sum_{i=1}^n \phi_i(\theta)$ be the sample mean. The GMM criterion is

$$Q_n(\theta) := n \bar{\phi}_n(\theta)' W_n \bar{\phi}_n(\theta),$$

where W_n is a positive-definite weighting matrix. We implement either two-step GMM, with $W_n = \hat{V}_{\phi\phi}(\hat{\theta}_1)^{-1}$ where $\hat{\theta}_1$ is a preliminary consistent estimator, or the continuously updated estimator (CUE), with $W_n = \hat{V}_{\phi\phi}(\theta)^{-1}$. The cluster-robust variance estimator,

$$\hat{V}_{\phi\phi}(\theta) := \frac{1}{n} \sum_{i=1}^n (\phi_i(\theta) - \bar{\phi}_n(\theta)) (\phi_i(\theta) - \bar{\phi}_n(\theta))',$$

is consistent for $n \rightarrow \infty$ with T fixed and accommodates arbitrary within-firm serial dependence.

3 Identification Analysis

This section studies identification of the production elasticities β after the control function, and when it is weak. The key object is the orthogonalized Jacobian Π , which records residual instrument variation for β once the nonparametric productivity dynamics have been projected out.

For intuition, combine the productivity transition and the proxy into a composite control $g \circ h$ of

lagged proxies. Treating $g \circ h$ as an unrestricted function of p_{it-1} gives the partially linear structure

$$y_{it} = \beta' x_{it} + (g \circ h)(p_{it-1}) + u_{it},$$

with $\mathbb{E}[u_{it} \mid x_{it}, p_{it-1}] = 0$. Using the insight of Robinson (1988), β is identified only if residual input variation after projecting on p_{it-1} has full rank,

$$\mathbb{E}[(x_{it} - \mathbb{E}[x_{it} \mid p_{it-1}])(x_{it} - \mathbb{E}[x_{it} \mid p_{it-1}])']$$

full column rank. If that matrix is rank-deficient, inputs are linearly absorbed by the nonparametric control and β is not identified. The rest of the section constructs this analogue Π and studies both exact and almost rank failure, the latter corresponding to weak identification.

3.1 Concept of Identification

The model is identified if there exists a unique $\theta_0 \in \Theta$ satisfying the moment conditions

$$\mathbb{E}[\phi_{it}(\theta_0)] = 0.$$

Global identification requires this θ_0 to be the unique solution over Θ ; we maintain uniqueness as a high-level assumption and focus on local identification strength.²

Jacobian structure in the control-function model. We focus on $\vartheta := (\beta', \rho')'$, since γ is generically identified from the first-stage linear regression.³ The second-stage moment vector is $m(\vartheta) := \mathbb{E}[w_{2,it} u_{2,it}(\vartheta)]$. Using the parametric second-stage equation (10), the gradient of $u_{2,it}$ with respect to ϑ is

$$q_{it}(\vartheta_0) := - \left. \frac{\partial u_{2,it}(\vartheta)}{\partial \vartheta} \right|_{\vartheta_0} = \begin{pmatrix} x_{it} - b(r'_{it-1} \lambda_0)' \rho_0 \cdot x_{it-1} \\ b(r'_{it-1} \lambda_0) \end{pmatrix}, \quad (12)$$

²See Hansen (1982); Newey and McFadden (1994) for primitive conditions. Local identification governs the curvature of the criterion surface and the precision of estimation.

³The first stage (9) regresses y_{it} on r_{it} , which consists entirely of predetermined functions of the proxy; all regressors are exogenous by Assumption 2, so identification is generally nonproblematic.

where $\dot{b}(u) := \partial b(u)/\partial u = (1, 2u, \dots, L_g u^{L_g-1})'$ and $r'_{it-1} \lambda_0 = h_{t-1}(p_{it-1})$ at true values by definition of λ_0 . The second-stage Jacobian is then

$$\mathcal{J}_2 = \mathbb{E}[w_{2,it} q_{it}(\vartheta_0)']. \quad (13)$$

Orthogonalized Jacobian and the identification rank condition. Local identification of (β, ρ) is governed by residual variation in $\mathcal{J}_2$ after projecting out γ . Let

$$\mathcal{J}_\gamma := \frac{\partial}{\partial \gamma'} \mathbb{E}[w_{2,it} u_{2,it}(\vartheta, \gamma)] \Big|_{\theta_0}$$

collect those nuisance columns. Following standard practice in the weak-identification literature, we measure moments in units of their sampling variability: let $V_{\phi\phi}^{-1/2}$ be the symmetric inverse square root of $V_{\phi\phi}$ and define the standardized Jacobians

$$\tilde{\mathcal{J}}_2 := V_{\phi\phi}^{-1/2} \mathcal{J}_2, \quad \tilde{\mathcal{J}}_\gamma := V_{\phi\phi}^{-1/2} \mathcal{J}_\gamma.$$

Let $M_\gamma := I - \tilde{\mathcal{J}}_\gamma (\tilde{\mathcal{J}}_\gamma' \tilde{\mathcal{J}}_\gamma)^{-1} \tilde{\mathcal{J}}_\gamma'$ denote the orthogonal projector onto the complement of $\text{col}(\tilde{\mathcal{J}}_\gamma)$.

The orthogonalized Jacobian is

$$\Pi := M_\gamma \tilde{\mathcal{J}}_2. \quad (14)$$

(β, ρ) is locally identified if and only if Π has full column rank $d_\beta + d_\rho$; equivalently, there is no $\delta \neq 0$ with $\mathcal{J}_2 \delta \in \text{col}(\mathcal{J}_\gamma)$. Because $V_{\phi\phi}^{-1/2}$ is nonsingular, standardization leaves this rank condition and the failures catalogued in Section 3.2 unchanged; it matters only for the measures of identification strength in Section 3.3, where it makes $\|\cdot\|$ the Euclidean norm on moments scaled by their standard deviations.

3.2 Identification Failure

Example: Cobb–Douglas and AR(1) law of motion. Output follows Cobb–Douglas,

$$y_{it} = \beta_k k_{it} + \beta_\ell \ell_{it} + \omega_{it} + \varepsilon_{it}, \quad (15)$$

with productivity AR(1),

$$\omega_{it} = \rho \omega_{it-1} + \xi_{it}. \quad (16)$$

The proxy is intermediate input m_{it} , with informativeness summarized by the linear conditional mean

$$\mathbb{E}[\omega_{it} \mid k_{it}, m_{it}] = \delta_k k_{it} + \delta_m m_{it}, \quad (17)$$

where δ_m indexes the marginal signal in m_{it} about ω_{it} . Under AR(1), the second-stage equation (10) specializes to

$$y_{it} = \alpha_2 + \beta_k k_{it} + \beta_\ell \ell_{it} + \rho(r'_{it-1} \gamma - \beta_k k_{it-1} - \beta_\ell \ell_{it-1}) + u_{2,it},$$

with instruments $w_{2,it} = (1, k_{it}, k_{it-1}, \ell_{it-1}, m_{it-1})'$. The second-stage Jacobian columns for $\vartheta = (\beta_k, \beta_\ell, \rho)'$ are

$$\begin{aligned} \mathcal{J}_{\beta_k} &= \mathbb{E}[w_{2,it}(k_{it} - \rho_0 k_{it-1})], \\ \mathcal{J}_{\beta_\ell} &= \mathbb{E}[w_{2,it}(\ell_{it} - \rho_0 \ell_{it-1})], \\ \mathcal{J}_\rho &= \mathbb{E}[w_{2,it}(r'_{it-1} \gamma_0 - \beta_{k0} k_{it-1} - \beta_{\ell 0} \ell_{it-1})] = \mathbb{E}[w_{2,it} \omega_{it-1}], \end{aligned}$$

where the last equality uses that the dynamic index recovers ω_{it-1} at true values, and

$$\mathcal{J}_\gamma = -\rho_0 \mathbb{E}[w_{2,it} r'_{it-1}].$$

Functional dependence. Under Cobb–Douglas technology, labor satisfies a linear policy in capital and productivity⁴:

$$\ell_{it} = c_0 + c_1 k_{it} + c_2 \omega_{it}. \quad (18)$$

Under AR(1) productivity, $\ell_{it} - \rho_0 \ell_{it-1} = c_1(k_{it} - \rho_0 k_{it-1}) + c_2 \xi_{it}$. Since $\mathbb{E}[\xi_{it} \mid w_{2,it}] = 0$ by Assumption 2, $\mathcal{J}_{\beta_\ell} = c_1 \mathcal{J}_{\beta_k}$, so $\mathcal{J}_2 \delta = 0 \in \text{col}(\mathcal{J}_\gamma)$ for $\delta = (-c_1, 1, 0)'$. Conditional on ω_{it-1} , instruments that move capital move labor one-for-one through the policy rule.

Limited proxy signal. Suppose *sluggish capital* ($k_{it} = k_{it-1}$)⁵ and *zero proxy signal* ($\delta_m = 0$). Then $\mathbb{E}[\omega_{it-1} \mid w_{2,it}] = \delta_k \mathbb{E}[k_{it-1} \mid w_{2,it}]$ and $\mathbb{E}[k_{it} - \rho_0 k_{it-1} \mid w_{2,it}] = (1 - \rho_0) \mathbb{E}[k_{it-1} \mid w_{2,it}]$, so $\mathcal{J}_\rho = \frac{\delta_k}{1 - \rho_0} \mathcal{J}_{\beta_k}$ and $\mathcal{J}_2 \delta = 0 \in \text{col}(\mathcal{J}_\gamma)$ for $\delta = (-\frac{\delta_k}{1 - \rho_0}, 0, 1)'$.

These conditions—exact functional dependence and a fully uninformative proxy—are knife-edges

⁴See Appendix B.1.

⁵LP's findings suggest that capital investment is often sluggish for a wide range of industries, lending empirical support to this assumption.

unlikely to hold literally. The structural forces that generate them are nonetheless present in many production settings, and when they hold approximately rather than exactly, weak identification problems arise. Section 3.3 makes this precise.

Sources of rank deficiency. By Section 3.1, Π is column-rank deficient precisely when $\mathcal{J}_2\delta \in \text{col}(\mathcal{J}_\gamma)$ for some $\delta \neq 0$. The example above exhibits two such δ , and each reflects a structural feature of the control-function model rather than an accident of the parameterization: one acts through the β -columns of $\mathcal{J}_2$, the other through its ρ -columns. We state each in turn.

(i) *Functional dependence.* The column of $\mathcal{J}_2$ associated with β_j is the instrumented quasi-difference of input j ,

$$\mathcal{J}_{\beta_j} = \mathbb{E}[w_{2,it} (x_{j,it} - \dot{b}(r'_{it-1}\lambda_0)' \rho_0 x_{j,it-1})].$$

Let the static inputs obey the policy rule $v_{it} = v(k_{it}, \omega_{it})$. Their quasi-differences are then functions of the quasi-differenced capital stock and of the productivity innovation alone, and the innovation does not contribute to $\mathcal{J}_{\beta_v}$ because $\mathbb{E}[\xi_{it} \mid w_{2,it}] = 0$ under Assumption 2. The static-input columns therefore reproduce the capital column up to the coefficients of the policy rule, so that

$$\sum_j a_j \mathcal{J}_{\beta_j} \in \text{col}(\mathcal{J}_\gamma) \quad \text{for some } a \neq 0,$$

and $\mathcal{J}_2\delta \in \text{col}(\mathcal{J}_\gamma)$ for the corresponding δ supported on the β coordinates. Under the exact policy rule (18) the sum vanishes identically.⁶

(ii) *Limited proxy signal.* The columns associated with ρ are

$$\mathcal{J}_\rho = \mathbb{E}[w_{2,it} b(r'_{it-1}\lambda_0)'],$$

whose argument $r'_{it-1}\lambda_0$ is measurable with respect to lagged inputs. Separating ρ from β thus requires the proxy index to move independently of the lagged inputs already entering the β -columns, and identification fails when it does not:

$$\text{col}(\mathcal{J}_\rho) \subseteq \text{span}\{\mathcal{J}_{\beta_j}\} + \text{col}(\mathcal{J}_\gamma),$$

⁶Ackerberg et al. (2015) locate functional dependence in the *first-stage* regression. The same policy rule operates here at the *second stage*, where it induces linear dependence among the β -columns of $\mathcal{J}_2$ relative to $\text{col}(\mathcal{J}_\gamma)$.

in which case $\mathcal{J}_2\delta \in \text{col}(\mathcal{J}_\gamma)$ for some $\delta \neq 0$ with a nonzero ρ coordinate. In the example this inclusion holds when the proxy carries no marginal signal, $\delta_m = 0$, and capital is sluggish, so that the index reduces to a function of lagged capital alone.

Remark 1 (Signal attenuation versus invertibility failure). “Weak” or “noisy” proxy here means signal attenuation: $\mathcal{J}_\rho$ approaches $\text{span}\{\mathcal{J}_{\beta_j}\} + \text{col}(\mathcal{J}_\gamma)$, so some $\delta \neq 0$ has $\mathcal{J}_2\delta$ approaching that space. This is distinct from the Levinsohn and Petrin (2003) invertibility failure, which precludes recovering ω_t . If instead $\eta_{it} := \omega_{it} - \mathbb{E}[\omega_{it} \mid p_{it}]$ remains correlated with flexible inputs, the moments are misspecified. Under the index-sufficiency condition of Section 3.5 (Doraszelski and Li, 2025), that correlation is ruled out and the channel is only one of identification strength.

3.3 Weak Identification

Neither condition of Section 3.2 is likely to hold exactly. Both hold approximately in many production settings, and approximate failure is already enough to invalidate conventional inference.

Sources of near-rank deficiency. Identification strength is summarized by the concentration matrix

$$\mathcal{C} := \Pi'\Pi = \tilde{\mathcal{J}}_2' M_\gamma \tilde{\mathcal{J}}_2. \quad (19)$$

For any direction $\delta \in \mathbb{R}^{d_\beta + d_\rho}$,

$$\delta' \mathcal{C} \delta = \|\Pi\delta\|^2 = \min_{c \in \mathbb{R}^{d_\gamma}} \|\tilde{\mathcal{J}}_2\delta - \tilde{\mathcal{J}}_\gamma c\|^2, \quad (20)$$

where $\|\cdot\|$ is the Euclidean norm and $d_\gamma = \dim(\gamma)$. The second equality is the least-squares characterization of the projection: the minimizing c is $(\tilde{\mathcal{J}}_\gamma' \tilde{\mathcal{J}}_\gamma)^{-1} \tilde{\mathcal{J}}_\gamma' \tilde{\mathcal{J}}_2\delta$, leaving the residual $M_\gamma \tilde{\mathcal{J}}_2\delta = \Pi\delta$, so (20) measures how far $\tilde{\mathcal{J}}_2\delta$ lies from the nuisance span $\text{col}(\tilde{\mathcal{J}}_\gamma)$. Thus $\|\Pi\delta\|$ is at once the identifying response of the standardized moments to a perturbation of ϑ along δ and the distance between the design at hand and the exact failure of Section 3.2, at which the minimum in (20) is zero. Let either condition of that section hold approximately, in the sense that $\delta' \mathcal{C} \delta$ is small but nonzero for some δ with $\|\delta\| = 1$, equivalently that $\lambda_{\min}(\mathcal{C})$ is small. Then Π retains full column rank and (β, ρ) remains point identified, while the moments respond to movements in ϑ along δ only at order $\|\Pi\delta\|$: the criterion is nearly flat in that direction, and the sampling distribution of the estimator departs from its normal approximation. Identification is weak rather than absent. The statistic

T_{CD} of Section 4.1 is the sample analogue of $\lambda_{\min}(n\mathcal{C})$, so (19) is the population object that the empirical diagnostics of Section 6 estimate. Both channels of Section 3.2 operate continuously in this metric. Frictions that break the static-input policy rule—optimization error, adjustment costs, heterogeneous input prices—leave the quasi-differenced flexible input with independent variation of the same order as the frictions themselves; proxy noise attenuates the variation in $\mathcal{J}_\rho$ that separates ρ from β , while leaving the moment conditions valid under the index-sufficiency condition of Section 3.5 (Remark 1). In each case a single structural quantity scales $\|\Pi\delta\|$, up to a geometric factor measuring how much of the resulting variation survives projection on the nuisance span, as the example makes explicit.

Example: Cobb–Douglas and AR(1) law of motion (continued). With optimization noise

$\ell_{it} = \ell_{it}^* + \xi_{it}^l$, $\xi_{it}^l \sim \mathcal{N}(0, \sigma_{\xi_l}^2)$, $\mathbb{E}[\xi_{it}^l \mid z_{i,t-1}] = 0$, and ℓ_{it}^* as in (18),

$$\mathcal{J}_{\beta_\ell} = c_1 \mathcal{J}_{\beta_k} - \rho_0 \mathbb{E}[w_{2,it} \xi_{it-1}^l],$$

where $\mathbb{E}[w_{2,it} \xi_{it-1}^l] = \sigma_{\xi_l}^2 \mathbf{e}_\ell$ and $\mathbf{e}_\ell$ selects the ℓ_{it-1} coordinate of $w_{2,it}$. Evaluating this along the failure direction $\delta = (-c_1, 1, 0)'$ of Section 3.2,

$$\mathcal{J}_2 \delta = -\rho_0 \sigma_{\xi_l}^2 \mathbf{e}_\ell, \quad \|\Pi\delta\| = |\rho_0| \sigma_{\xi_l}^2 \kappa_\ell, \quad \kappa_\ell := \|M_\gamma V_{\phi\phi}^{-1/2} \mathbf{e}_\ell\|.$$

Because $\|\delta\|^2 = 1 + c_1^2$, combining this with (20) and the variational characterization of the smallest eigenvalue bounds the concentration matrix:

$$\lambda_{\min}(\mathcal{C}) \leq \frac{\delta' \mathcal{C} \delta}{\|\delta\|^2} = \frac{\rho_0^2 \kappa_\ell^2}{1 + c_1^2} \sigma_{\xi_l}^4. \quad (21)$$

Labor optimization noise is thus the quantity that indexes the distance to exact failure along δ : the identifying response $\|\Pi\delta\|$ is of order $\sigma_{\xi_l}^2$, so the concentration quadratic form is of order $\sigma_{\xi_l}^4$. At $\sigma_{\xi_l} = 0$ the knife-edge of Section 3.2 obtains and β_ℓ is not identified; for $\sigma_{\xi_l} > 0$ it is identified, with strength governed by σ_{ξ_l} at these rates. Small but nonzero optimization noise therefore delivers precisely the weakly identified case. Two qualifications matter for reading (21). First, $V_{\phi\phi}$ and M_γ are themselves functions of σ_{ξ_l} , so κ_ℓ is not a constant; the stated orders are leading-order statements as $\sigma_{\xi_l} \rightarrow 0$, valid because κ_ℓ tends to a finite nonzero limit whenever $V_{\phi\phi}^{-1/2} \mathbf{e}_\ell$ does not lie in $\text{col}(\tilde{\mathcal{J}}_\gamma)$

in that limit. Second, strength is the product of a structural magnitude and the geometric factor κ_ℓ : identification is also weak when κ_ℓ is small with σ_{ξ_t} bounded away from zero, that is, when the residual variation in ℓ_{it-1} is nearly absorbed by the nuisance span. The Monte Carlo of Section 5 operates in this regime, holding σ_{ξ_t} small while varying the proxy noise.

With classical proxy error $m_{it} = m_{it}^* + \nu_{it}$, $\mathbb{E}[\nu_{it} \mid z_{i,t-1}] = 0$,

$$\mathbb{E}[\omega_{it-1} \mid k_{it-1}, m_{it-1}] = \delta_k k_{it-1} + \tilde{\delta}_m(\sigma_\nu) m_{it-1},$$

where $\tilde{\delta}_m(\sigma_\nu) := \delta_m \cdot \sigma_{m^*}^2 / (\sigma_{m^*}^2 + \sigma_\nu^2)$.⁷ Under sluggish capital,

$$\mathcal{J}_\rho = \frac{\delta_k}{1 - \rho_0} \mathcal{J}_{\beta_k} + \tilde{\delta}_m(\sigma_\nu) \mathbb{E}[w_{2,it} m_{it-1}].$$

Evaluating this along the failure direction $\delta = (-\frac{\delta_k}{1-\rho_0}, 0, 1)'$ of Section 3.2,

$$\mathcal{J}_2 \delta = \tilde{\delta}_m(\sigma_\nu) \mathbb{E}[w_{2,it} m_{it-1}], \quad \|\Pi \delta\| = \tilde{\delta}_m(\sigma_\nu) \kappa_m, \quad \kappa_m := \|M_\gamma V_{\phi\phi}^{-1/2} \mathbb{E}[w_{2,it} m_{it-1}]\|,$$

and, since $\|\delta\|^2 = 1 + \delta_k^2 / (1 - \rho_0)^2$,

$$\lambda_{\min}(\mathcal{C}) \leq \frac{\kappa_m^2}{1 + \delta_k^2 / (1 - \rho_0)^2} \tilde{\delta}_m(\sigma_\nu)^2.$$

The attenuation factor $\tilde{\delta}_m(\sigma_\nu)$ therefore plays the role here that $\sigma_{\xi_t}^2$ plays above, with the same two qualifications. Since $\tilde{\delta}_m$ is strictly decreasing in σ_ν , identification weakens monotonically in proxy noise and fails only in the limit $\sigma_\nu \rightarrow \infty$; for finite σ_ν , identification of f can survive under Section 3.5.

Remark 2 (Local-to-zero nesting). These two expressions make the drifting-Jacobian device of Section 3.4 concrete. Taking the coordinate along δ as the weakly identified direction ϑ_w , the local-to-zero condition $M_{w,n} = n^{-1/2}C$ of Assumption 4 holds along sequences of designs in which $\sigma_{\xi_t}^2$ —equivalently $\sigma_{\xi_t} \propto n^{-1/4}$ —or, for the proxy channel, $\tilde{\delta}_m(\sigma_\nu)$, shrinks at rate $n^{-1/2}$. Weak identification is therefore not a property of the estimator but a statement about how close a design sits to the failures of Section 3.2.

⁷Standard attenuation: $\sigma_{m^*}^2 = \text{Var}(m_{it}^*)$.

3.4 Asymptotic theory

When Π is nearly rank-deficient, the GMM criterion is nearly flat along the corresponding directions in ϑ , the estimator has a nonstandard sampling distribution, and conventional inference is invalid.

We now make this precise; proofs are in Appendix A.

Let $\vartheta = (\vartheta'_w, \vartheta'_s)' \in \Theta_w \times \Theta_s$, where $\vartheta_w \in \mathbb{R}^{d_w}$ collects the weakly identified directions—those along which Π is nearly singular—and $\vartheta_s \in \mathbb{R}^{d_s}$ the strongly identified directions, with $d_w + d_s = d_\beta + d_\rho$. Any subset of the structural parameters may belong to either block. We show that the asymptotic behaviour of the entire parameter ϑ is affected even if the Jacobian is only partially near-rank reduced.

Define the population moments $m_n(\vartheta_w, \vartheta_s) := \mathbb{E}[\bar{\phi}_n(\vartheta_w, \vartheta_s)]$. Using a first-order expansion around $\vartheta_0 = (\vartheta'_{w0}, \vartheta'_{s0})'$,

$$m_n(\vartheta_w, \vartheta_s) = m_n(\vartheta_{w0}, \vartheta_{s0}) + M_{w,n}(\vartheta_w, \vartheta_s)(\vartheta_w - \vartheta_{w0}) + M_{s,n}(\vartheta_w, \vartheta_s)(\vartheta_s - \vartheta_{s0}) + r_n(\vartheta_w, \vartheta_s), \quad (22)$$

where

$$M_{w,n}(\vartheta_w, \vartheta_s) := \mathbb{E}\left[\frac{\partial \bar{\phi}_n(\vartheta_w, \vartheta_s)}{\partial \vartheta'_w}\right], \quad M_{s,n}(\vartheta_w, \vartheta_s) := \mathbb{E}\left[\frac{\partial \bar{\phi}_n(\vartheta_w, \vartheta_s)}{\partial \vartheta'_s}\right].$$

We model a partially nearly rank deficient Jacobian following Stock and Wright (2000) as a sequence of DGPs along which $M_{w,n} = O(n^{-1/2})$ while $M_{s,n}$ remains of fixed order.

Assumption 4 (Partial local-to-zero Jacobian). Along a sequence of DGPs indexed by n ,

- (i) $M_{w,n}(\vartheta_w, \vartheta_s) = n^{-1/2}C(\vartheta_w, \vartheta_s)$, where $C(\vartheta_w, \vartheta_s) \in \mathbb{R}^{d_\phi \times d_w}$ is continuous in ϑ_s and bounded on $\Theta_w \times \Theta_s$.
- (ii) $M_{s,n}(\vartheta_w, \vartheta_s) = R(\vartheta_w)$, where $R(\vartheta_w) \in \mathbb{R}^{d_\phi \times d_s}$ is continuous and bounded on Θ_w .
- (iii) $[C(\vartheta_w, \vartheta_s) \ R(\vartheta_w)]$ has full column rank uniformly on $\Theta_w \times \Theta_s$.
- (iv) $\sup_{(\vartheta_w, \vartheta_s) \in \Theta_w \times \Theta_s} \|r_n(\vartheta_w, \vartheta_s)\| = o(n^{-1/2})$.

Condition (i) formalizes the near rank-reduced Jacobian that vanishes at rate $n^{-1/2}$ in the weakly identified directions. Condition (ii) requires that the strongly identified directions retain an $O(1)$ Jacobian, so those parameters remain well-identified along the sequence. Condition (iii) ensures the problem is *partially* weakly identified: the Jacobian has full column rank, so the model is not fully

unidentified. Condition (iv) requires the remainder in the first-order expansion to be $o(n^{-1/2})$, so the linear approximation is exact at the scale of the weak Jacobian.

Assumption 4 extends Stock and Wright (2000) in one respect: the strong-block Jacobian $R(\vartheta_w)$ is allowed to depend on the weakly identified parameters. Stock and Wright (2000) require the strong-block moment function to be independent of the weak block, so their analogue of R is a fixed matrix evaluated at the true parameter value. In the production function model, however, $R(\vartheta_w) = \mathbb{E}[w_{2,it} b(r'_{it-1}\gamma - x'_{it-1}\beta)']$ depends on $\beta = \vartheta_w$, so that assumption is inapplicable. The consequence is that consistency of $\hat{\vartheta}_s$ cannot be established via a fixed population criterion: the separation argument in Proposition 7 must hold uniformly over Θ_w , which requires the uniform full-column-rank condition on $R(\vartheta_w)$ in condition (iii).

Although the estimator might have non-standard distribution, the sample moments on which it is based are asymptotically uniformly Gaussian under weak conditions. Assumption 5 requires a functional CLT for the empirical process; it is implied by standard primitive conditions such as Lipschitz moment functions with square-integrable Lipschitz constants (Andrews, 1994).

Assumption 5 (Uniform Gaussian limit). The centered empirical process $\psi_n(\vartheta) := \sqrt{n}\{\bar{\phi}_n(\vartheta) - m_n(\vartheta)\}$ converges weakly on $\Theta_w \times \Theta_s$ to a mean-zero Gaussian process ψ_0^f with almost surely bounded sample paths.

Assumption 6 is a standard regularity condition on the weighting matrix.

Assumption 6 (Positive definite weighting matrix). $W_n \xrightarrow{p} W_0$ with W_0 symmetric positive definite.

Proposition 7 ($\sqrt{n}$ -consistency of the strongly identified block). *Under Assumptions 4–6, for any GMM minimizer $\hat{\vartheta} = (\hat{\vartheta}'_w, \hat{\vartheta}'_s)'$ of $Q_n(\vartheta) = n \bar{\phi}_n(\vartheta)' W_n \bar{\phi}_n(\vartheta)$:*

$$\hat{\vartheta}_s \xrightarrow{p} \vartheta_{s0} \quad \text{and} \quad \sqrt{n}(\hat{\vartheta}_s - \vartheta_{s0}) = O_p(1).$$

Since Proposition 7 establishes $\sqrt{n}$ -consistency of $\hat{\vartheta}_s$, the behavior of the criterion along weakly identified directions is governed by $\sqrt{n}$ -local neighborhoods of ϑ_{s0} . It therefore suffices to study Q_n at perturbations of the form $\tilde{\vartheta}_s = \vartheta_{s0} + \lambda/\sqrt{n}$ for $\lambda \in \mathbb{R}^{d_s}$, so that $\sqrt{n}(\tilde{\vartheta}_s - \vartheta_{s0}) = O(1)$.

Proposition 8 (Uniform boundedness of the criterion). *Under Assumptions 4–6:*

$$\sup_{\vartheta_w \in \Theta_w} Q_n(\vartheta_w, \tilde{\vartheta}_s) = O_p(1).$$

Proposition 8 is the analogue of Stock and Wright (2000, Theorem 1) for the partially weakly identified GMM setting. Because $Q_n(\vartheta_w, \tilde{\vartheta}_s)$ is stochastically bounded uniformly over Θ_w , the weakly identified component $\hat{\vartheta}_w = O_p(1)$ has a generally non-normal limiting distribution. The strongly identified block remains consistent, but its limiting distribution is also generally non-normal because it depends on the random limit of $\hat{\vartheta}_w$. Conventional Wald inference is therefore unreliable, which motivates the identification-robust procedures in Section 4.

3.5 Nuisance-Function Misspecification

Assumption 3 imposes exact invertibility: $\omega_{it} = h_t(p_{it})$. In practice, measurement error, optimization noise, or input-price heterogeneity cause the proxy to recover productivity only partially. This misspecifies the canonical nuisance structure, but need not preclude identification of f .

Replace Assumption 3 with the conditional mean condition

$$\mathbb{E}[\omega_{it} \mid p_{it}] = h_t(p_{it}), \quad \eta_{it} := \omega_{it} - h_t(p_{it}), \quad \mathbb{E}[\eta_{it} \mid p_{it}] = 0. \quad (23)$$

Under exact invertibility, the canonical law of motion gives $\mathbb{E}[\omega_{it} \mid p_{it-1}] = g(h_{t-1}(p_{it-1}))$. Under (23), ω_{it-1} is no longer exactly recovered from p_{it-1} , so this equality fails and g ceases to be directly estimable. Write $\eta_{it} := \omega_{it} - h_t(p_{it})$ as in (23). If flexible inputs remain correlated with η_{it} , the residual is a source of endogeneity after the control and the second-stage moments fail. Doraszelski and Li (2025) show that identification is nonetheless preserved when the conditional mean of productivity given the proxy index satisfies an index-sufficiency condition, which makes the residual orthogonal to the relevant information set—the case in which noisy proxies weaken Π without invalidating the moments (Remark 1).

Assumption 9 (Index sufficiency). There exists a link function $\tilde{g}(\cdot)$ such that $\mathbb{E}[\omega_{it} \mid p_{it-1}] = \tilde{g}(h_{t-1}(p_{it-1}))$.

$\tilde{g}$ is a reduced-form object: it characterizes the conditional mean of productivity given the proxy index, rather than the structural transition g . Substituting Assumption 9 into the production function gives

$$y_{it} = f(k_{it}, v_{it}) + \tilde{g}(h_{t-1}(p_{it-1})) + u_{2,it}^*,$$

where $u_{2,it}^* := \nu_{it-1} + \xi_{it} + \varepsilon_{it}$ and $\nu_{it-1} := g(\omega_{it-1}) - \tilde{g}(h_{t-1}(p_{it-1}))$ satisfies $\mathbb{E}[\nu_{it-1} \mid p_{it-1}] = 0$

by Assumption 9. Under nuisance-function misspecification, the canonical structural decomposition fails, but the second-stage moment conditions $\mathbb{E}[w_{2,it} u_{2,it}^*] = 0$ still hold for the modified composite error, so f remains identified.

The limited-proxy-signal channel of Section 3.3 is therefore best understood as degradation of identification *strength* rather than identification failure: as proxy noise grows under index sufficiency, the Jacobian weakens continuously, producing the near-rank deficiency analyzed above, while formal identification is preserved. That conclusion is complementary to the functional-dependence channel: both operate by shrinking residual variation in Π , and neither requires abandoning identification-robust inference in favor of a different identification strategy.

4 Identification Robust Methods

We employ identification-robust inference procedures for the production function parameters within the control-function approach. Conventional inference methods, such as Wald statistics and t -statistics, rely on asymptotic normality, which breaks down under weak identification, resulting in nonstandard distributions and size distortions (Staiger and Stock, 1997; Stock and Wright, 2000). Selecting instruments or proxy variables based on standard pre-testing, such as the commonly used F -test, does not resolve the issue, as it may introduce selection bias resulting in a test that is not size-controlled.

Identification-robust statistics overcome these problems by maintaining a valid limiting distribution regardless of identification strength, ensuring reliable hypothesis testing and confidence intervals (Andrews and Stock, 2005). These statistics are defined for the CUE.

4.1 Rank test for identification

A straightforward approach to diagnosing identification strength is to directly test whether the Jacobian $\mathcal{J}$ has full column rank. Since γ is generically well-identified from the first stage, the identification question of interest concerns the orthogonalized Jacobian Π defined in (14): rank deficiency of Π signals that (β, ρ) cannot be separately recovered from $g(\cdot)$. In practice we implement the test on the full second-stage Jacobian $\widehat{\mathcal{J}}_{2,n}$ after concentrating out the estimated nuisance directions, which yields a direct test of $\text{rank}(\Pi) = d_\beta + d_\rho$.

The null hypothesis is that the (orthogonalized) Jacobian has exactly one rank deficiency:

$$H_0 : \text{rank}(\mathcal{J}) = d_\theta - 1,$$

against the alternative that $\text{rank}(\mathcal{J}) = d_\theta$. We test this using the minimum eigenvalue of the standardized Jacobian cross-product, the full-parameter counterpart of the concentration matrix (19), evaluated at the CUE estimate $\hat{\theta}$:

$$T_{\text{CD}} = \lambda_{\min}\left(n \hat{\mathcal{J}}_n(\hat{\theta})' \hat{V}_{\phi\phi}(\hat{\theta})^{-1} \hat{\mathcal{J}}_n(\hat{\theta})\right), \quad (24)$$

where $\hat{\mathcal{J}}_n(\hat{\theta}) = \partial \bar{\phi}_n(\theta) / \partial \theta' |_{\hat{\theta}}$. This is the nonlinear GMM analogue of the Cragg–Donald minimum-eigenvalue statistic (Cragg and Donald, 1993). Under H_0 , $T_{\text{CD}} \xrightarrow{d} \chi_{d_\phi - d_\theta + 1}^2$. Large values reject H_0 and signal identification; failure to reject signals potential underidentification. In the just-identified case ($d_\phi = d_\theta$) the degrees of freedom equal one.

In nonlinear GMM, $\hat{\mathcal{J}}_n(\theta)$ depends on θ and $\hat{\theta}$ is inconsistent under H_0 . Wright (2003) shows that the formally valid procedure eliminates θ by minimizing $T_{\text{CD}}(\theta)$ over the Stock–Wright S-set, yielding a statistic $L^* = \inf_{\theta \in S^*(\alpha)} T_{\text{CD}}(\theta)$ that is asymptotically conservative with size $\leq 2\alpha$. Since $L^* \leq T_{\text{CD}}$ by construction, our diagnostic is *favourable* towards finding identification: whenever T_{CD} rejects H_0 , so does L^* . Failure to reject T_{CD} therefore constitutes strong evidence of underidentification.

Beyond the minimum eigenvalue, the full set of eigenvalues of $n \hat{\mathcal{J}}_n(\hat{\theta})' \hat{V}_{\phi\phi}(\hat{\theta})^{-1} \hat{\mathcal{J}}_n(\hat{\theta})$ — one per parameter — is informative: when identification is strong all eigenvalues diverge, while a near-zero eigenvalue in a particular direction reveals which linear combination of parameters is poorly identified.

4.2 Conditional first-stage F as a comparative diagnostic

The eigenvalue spectrum of the concentration matrix points to a weak direction in parameter space, but not to the input whose instrument variation the control has absorbed. To localize that, we adapt the conditional first-stage F of Sanderson and Windmeijer (2016) to the β -block of the second-stage score (12), whose quasi-differenced inputs

$$\tilde{x}_{it}(\vartheta) := x_{it} - (\dot{b}(r'_{it-1}\lambda)' \rho) x_{it-1} \quad (25)$$

satisfy $\mathcal{J}_\beta = \mathbb{E}[w_{2,it} \tilde{x}_{it}(\vartheta_0)']$ and reduce to $x_{it} - \rho x_{it-1}$ under AR(1).

Evaluate $\tilde{x}_{it}(\vartheta)$ at the CUE and project it on $w_{2,it}$. For each coordinate j , residualize $\tilde{x}_{it,j}$ on the fitted values of the remaining coordinates and let F_j be the firm-cluster-robust Wald statistic for the relevance of $w_{2,it}$ in that residual.⁸ A near-zero F_j says that the apparent first-stage strength in input j is shared with the other inputs and disappears once that dependence is removed—the sample counterpart of the near-functional dependence in Section 3.2. Sanderson–Windmeijer critical values are calibrated for linear IV, whereas the Jacobian here is nonlinear and θ -dependent; we therefore read $\{F_j\}$ comparatively—across samples, or along a profile in ρ —rather than against Stock–Yogo thresholds.

4.3 Bootstrap pre-test for proxy relevance

While the rank test assesses identification at the level of the population Jacobian, it does not directly reveal whether the resulting distortion to the estimator’s sampling distribution is empirically consequential. Angelini et al. (2024) propose a bootstrap normality diagnostic that addresses this gap. Rather than examining the Jacobian, it tests whether the CUE’s finite-sample distribution conforms to the asymptotic normal approximation required for conventional inference to be valid—the key advantage being that it is symptom-based and therefore sensitive to the full downstream effect of weak identification, including nonlinearities that a Jacobian rank test alone may not capture. The null hypothesis is that all proxies (and instruments) are strong.

Let $\hat{\theta}_n$ denote the CUE obtained from the full sample and let V_θ be a consistent estimate of its asymptotic covariance, which we can compute from $\hat{V}_{\phi\phi}(\hat{\theta})$. Let $\hat{\theta}_n^*$ be the bootstrapped counterpart, using the Moving Block Bootstrap (MBB) technique. The MBB is similar in spirit to a standard bootstrap with replacement, but instead of resampling one observation at a time it resamples blocks of observations—in our case at the firm level—in order to replicate their serial dependence structure (Angelini et al., 2024). We can then form

$$\Gamma_n^* = \sqrt{n} V_\theta^{-1/2} (\hat{\theta}_n^* - \hat{\theta}_n).$$

Under the null, Γ_n^* is asymptotically standard normal. Testing multivariate normality—e.g. with the Doornik–Hansen or a set of Kolmogorov–Smirnov statistics—therefore provides a valid decision

⁸Coefficients on instruments that enter $\tilde{x}_{it,j}(\vartheta)$ by construction are excluded from the tested restriction so that F_j is not mechanical.

rule. Failure to reject normality implies that standard Wald inference remains appropriate; rejection signals weak proxies, in which case one should switch to identification-robust procedures such as the AR-GMM confidence sets introduced below. Proposition 8 is what gives the diagnostic its power: weak proxies deliver a nonregular, non-Gaussian limit for $\hat{\theta}_n$, so the studentized bootstrap increment can remain Gaussian only under strong identification.

The pre-test does not distort subsequent inference. Under the null of strong proxies, Γ_n^* has a nonrandom asymptotic law conditional on the sample, so the diagnostic decision is asymptotically independent of any statistic computed from the original data. Conditioning on nonrejection therefore leaves asymptotic rejection probabilities and coverage unchanged.

Implementation requires only the bootstrap re-estimations—no extra first-stage regressions, no critical-value adjustments—and accommodates multiple proxies, conditional heteroskedasticity, and zero-censored instruments. In practice we report all three diagnostics: the rank test and the conditional F -statistics assess Π jointly and componentwise, while the bootstrap pre-test establishes whether the distortion reaches the estimator’s sampling distribution.

4.4 AR-GMM test

A straightforward identification-robust statistic is the GMM extension of the AR test (Anderson and Rubin, 1949), the AR-GMM. It is based on the S-statistic of Stock and Wright (2000). Let $\theta \in \mathbb{R}^{d_\theta}$ be the parameter of interest and $\bar{\phi}_n(\theta) \in \mathbb{R}^{d_\phi}$ the sample moments with which it is estimated. To test the null hypothesis $H_0 : \theta_0 = \theta^*$, the S-statistic is equivalent to the CUE objective function evaluated at θ^* :

$$S(\theta^*) = n \bar{\phi}_n(\theta^*)' \widehat{V}_{\phi\phi}(\theta^*)^{-1} \bar{\phi}_n(\theta^*). \quad (26)$$

The S-statistic converges under H_0 to a $\chi_{d_\phi}^2$, irrespective of identification strength (see Theorem 2 in Stock and Wright, 2000).

The AR-GMM test has power against the alternative $\theta \neq \theta^*$ and against violations of moment restrictions in overidentified models. Its power diminishes as the number of moment restrictions increases, since the degrees of freedom of its limiting distribution equal the number of moment conditions. Consequently, when the number of instruments substantially exceeds the number of structural parameters, the statistic exhibits low power. To address this limitation, alternative,

more powerful identification-robust test statistics have been developed (Moreira, 2003; Kleibergen, 2002, 2005). We implement the KLM statistic of Kleibergen (2002) in its GMM form and the weak-identification-robust J -test of Kleibergen (2005) in our empirical applications; see Sections 4.5 and 4.5.1 below.

4.4.1 Robust Confidence Sets

From Proposition 2 it follows that the $100 \times (1 - \alpha)\%$ confidence set for θ is

$$CS(\alpha) = \{\theta^* : S(\theta^*) \leq \chi_{d_\phi}^2(1 - \alpha)\}, \quad (27)$$

where $\chi_{d_\phi}^2(1 - \alpha)$ denotes the $(1 - \alpha)$ quantile of the $\chi_{d_\phi}^2$ distribution. Since $S(\theta^*)$ is not a quadratic function of θ^* , the confidence set cannot be expressed as an estimator $\pm$ a multiple of a standard error. Instead, one must evaluate $S(\theta^*)$ on a grid of θ^* values and retain those that satisfy the inequality.

The shape of $CS(\alpha)$ is informative about identification. Bounded convex sets indicate strong identification. Bounded but non-convex sets may still signal identification difficulties. Unbounded sets arise under weak identification, when the criterion is nearly flat in some direction. Empty sets indicate that no parameter value satisfies the moment conditions, pointing to model misspecification. Wang and Zivot (1998) discuss these shapes in the bivariate AR case: elliptical sets correspond to strong identification, while hyperbolic sets indicate potential weak identification.

Confidence sets for a subvector of θ can be obtained by substituting the remaining elements with their estimates, then computing $S(\theta^*)$ over a grid varying only the subvector of interest. An alternative is to project a valid joint confidence set for θ onto the subspace of interest (Dufour, 1997, Sec. 5.2), which yields an asymptotically conservative confidence set.

4.4.2 AR-GMM tests on linear transformations of the parameter vector

In many applications the objects of interest are linear combinations of the structural parameters rather than the parameters themselves. Let $\theta \in \mathbb{R}^{d_\theta}$ denote the original parameter vector and

$$\psi := R\theta \in \mathbb{R}^{d_\psi}, \quad d_\psi \leq d_\theta,$$

be a parameter of interest defined by a known matrix R of full row rank. The remaining components of θ are treated as nuisance. Any one-to-one linear reparameterization of θ leaves the S-statistic and its limiting distribution invariant, so inference on ψ can be carried out by working directly with the original GMM objective.⁹

Write $\theta = (\psi', \eta')'$ after a suitable reparameterization, where $\eta \in \mathbb{R}^{d_\eta}$ with $d_\eta = d_\theta - d_\psi$ collects the nuisance components. To test $H_0 : \psi = \psi^*$, define the *restricted* CUE estimator of η as

$$\hat{\eta}(\psi^*) := \arg \min_{\eta} S(\psi, \eta) \big|_{\psi=\psi^*}.$$

The corresponding linear-transformation S-statistic is $S(\psi^*, \hat{\eta}(\psi^*))$.

Under H_0 , the limiting null distribution of $S(\psi^*)$ is stochastically dominated by a $\chi^2_{d_\phi - d_\eta}$, with equality when η is well identified.¹⁰ Using the $\chi^2_{d_\phi - d_\eta}(1 - \alpha)$ critical value therefore yields an asymptotically size-correct test that remains valid irrespective of the identification strength of η .

An identification-robust $(1 - \alpha) \times 100\%$ confidence set for ψ is obtained by inversion:

$$CS_\psi(\alpha) := \left\{ \psi \in \mathbb{R}^{d_\psi} : S(\psi, \hat{\eta}(\psi)) \leq \chi^2_{d_\phi - d_\eta}(1 - \alpha) \right\}.$$

In practice, computation proceeds by gridding over ψ , solving the restricted CUE problem for $\hat{\eta}(\psi)$ at each grid point, and retaining all values that satisfy the inequality. As in the full-parameter case, the resulting confidence set may be bounded, unbounded, or disconnected in the presence of weak identification.

Subset confidence sets are obtained by taking $\psi = \theta_1$ and $\eta = \theta_2$ in the partition $\theta = (\theta'_1, \theta'_2)'$ with $R = (I_{d_{\theta_1}}, 0)$. Subset AR-GMM procedures are (weakly) more powerful than projection-based methods: non-rejection by the linear-transformation test implies non-rejection by the test obtained from projecting the full-parameter AR-GMM confidence set onto the ψ -space, so projection-based sets are typically more conservative (Dufour, 1997).

4.5 Alternative identification-robust tests

The AR-GMM statistic is fully identification-robust but can have limited power in overidentified designs, since its limiting distribution is $\chi^2_{d_\phi}$ regardless of identification strength: more moment

⁹See, for example, Kleibergen and Mavroeidis (2009a) for a detailed discussion in linear IV and GMM settings.

¹⁰Formally, $S(\psi^*) \stackrel{d}{\Rightarrow} Q$, where Q is bounded above in the convex order by $\chi^2_{d_\phi - d_\eta}$ uniformly over parameter sequences that allow for weak identification; see Kleibergen and Mavroeidis (2009a, Thm. 2).

conditions mean more degrees of freedom and a higher critical value even when the extra moments contain little information about θ . Kleibergen (2002) proposes a more powerful alternative for linear instrumental variables regression, the K (or KLM) statistic, which exploits the Jacobian to sharpen inference while retaining a valid limiting distribution under weak identification; Kleibergen (2005) extends it to nonlinear GMM, which is the form we use.

The K statistic. The standard GMM LM statistic of Newey and West (1987) uses the sample Jacobian $q_T(\theta_0, Y) = \partial f_T(\theta_0, Y)/\partial \theta'$ as an estimator of the expected Jacobian $J_\theta(\theta_0)$. The key problem is that the limiting behavior of $q_T(\theta_0, Y)$ is *not* independent of the limiting behavior of the moments $f_T(\theta_0, Y)$ when $J_\theta(\theta_0)$ is weak or zero, so the LM statistic's limiting distribution depends on nuisance parameters under weak identification.

Kleibergen (2005) replaces q_T with an estimator $\hat{D}_T(\theta_0, Y)$ constructed from the first-order derivative of the CUE objective function. Specifically, the CUE first-order condition implies

$$\left. \frac{1}{2} \frac{\partial Q(\theta)}{\partial \theta'} \right|_{\theta_0} = f_T(\theta_0, Y)' \hat{V}_{ff}(\theta_0)^{-1} \hat{D}_T(\theta_0, Y),$$

where

$$\hat{D}_T(\theta_0, Y) = \left[q_{1,T}(\theta_0, Y) - \hat{V}_{\theta f,1}(\theta_0) \hat{V}_{ff}(\theta_0)^{-1} f_T(\theta_0, Y), \dots \right],$$

with $\hat{V}_{\theta f,i}(\theta_0)$ estimating the cross-covariance between the i -th column of q_T and f_T . The subtraction of $\hat{V}_{\theta f} \hat{V}_{ff}^{-1} f_T$ orthogonalizes the Jacobian estimator with respect to the moments. By Lemma 1 of Kleibergen (2005), $\hat{D}_T(\theta_0, Y)$ converges in distribution to a limit that is *independent* of the limit of $f_T(\theta_0, Y)$, regardless of whether $J_\theta(\theta_0)$ has full rank, is weak, or is zero. This independence is the key property that gives the K statistic its robustness.

The K statistic for $H_0 : \theta = \theta_0$ is then (Definition 1, Kleibergen, 2005):

$$K(\theta_0) = \frac{1}{T} f_T(\theta_0, Y)' \hat{V}_{ff}(\theta_0)^{-1/2} P_{\hat{V}_{ff}(\theta_0)^{-1/2} \hat{D}_T(\theta_0, Y)} \hat{V}_{ff}(\theta_0)^{-1/2} f_T(\theta_0, Y), \quad (28)$$

where $P_A = A(A'A)^{-1}A'$ denotes the projection onto the column space of A . Equivalently, writing $u = \sqrt{T} \hat{V}_{ff}^{-1/2} \bar{f}$ and $A = \hat{V}_{ff}^{-1/2} \hat{D}_T$, we have $K(\theta_0) = u' P_A u$. Since $\hat{D}_T$ is asymptotically independent of f_T , the matrix P_A is asymptotically independent of u , and the projection $u' P_A u$ converges to a $\chi_{d_\theta}^2$ distribution under H_0 irrespective of identification strength (Kleibergen, 2005, Theorem 1). Under strong identification, $K(\theta_0)$ is asymptotically equivalent to the standard GMM LM statistic

and therefore has higher power than the S-statistic when there are many overidentifying restrictions.

Subset K for individual parameters and returns to scale. In our application we are often interested in testing a hypothesis on a subvector of θ . Partition $\theta = (\alpha', \beta')'$ where β is the parameter of interest (e.g. a single elasticity or the sum of elasticities for RTS) and α collects the nuisance parameters. Following Kleibergen (2005), Section 3.2, the subset K statistic for $H_0 : \beta = \beta_0$ is obtained by projecting u and the columns of A corresponding to β onto the orthogonal complement of the columns of A corresponding to α :

$$K_\beta(\beta_0) = \tilde{u}' P_{\tilde{A}_\beta} \tilde{u},$$

where $\tilde{u} = M_{A_\alpha} u$, $\tilde{A}_\beta = M_{A_\alpha} A_\beta$, and $M_{A_\alpha} = I - P_{A_\alpha}$ is the annihilator of A_α . This partialling out removes the contribution of the nuisance directions and ensures that K_β converges to $\chi_{d_\beta}^2$ under H_0 and Assumption 3 of Kleibergen (2005) (that the nuisance Jacobian J_α has full rank). Confidence sets for β are obtained by inverting the test over a grid of β_0 values. For RTS, the constraint $\phi = \sum_v \beta_v$ defines a one-dimensional subspace, and we invert the subset K statistic over a grid of ϕ_0 values while profiling out the remaining elasticities.

Spurious power decline and the J statistic. The K statistic has one important limitation. Because K is a quadratic form in the first-order derivative of the CUE objective function $Q(\theta)$, it equals zero wherever $Q(\theta)$ is stationary: at its minimum (the CUE), at any local maximum, and at inflection points. At the CUE the moment conditions are (approximately) satisfied, so $K = 0$ there is correct. At a local maximum or inflection point, however, the moment conditions are *violated*—these are values of θ at which the data are inconsistent with the model—yet K still equals zero, falsely failing to reject H_0 at those points. This is the “spurious power decline” of Kleibergen (2005), Section 5: the K confidence set may include regions that are spurious because the moment conditions do not hold there.

The S-statistic $S(\theta_0) = (1/T) Q(\theta_0)$ is zero only at the CUE. Since $S = K + J$ (Kleibergen, 2005, eq. 30), where

$$J(\theta_0) = \frac{1}{T} f_T(\theta_0, Y)' \hat{V}_{ff}(\theta_0)^{-1/2} M_{\hat{V}_{ff}(\theta_0)^{-1/2} \hat{D}_T(\theta_0, Y)} \hat{V}_{ff}(\theta_0)^{-1/2} f_T(\theta_0, Y), \quad (29)$$

the J statistic captures the part of the criterion not explained by the K direction. Where K is spuriously small—at maxima and inflection points of Q —J is correspondingly large, because S is large at those points. Under H_0 and correct specification, $J(\theta_0) \xrightarrow{d} \chi_{d_\phi - d_\theta}^2$ independently of $K(\theta_0)$ (Kleibergen, 2005, Theorem, p. 1112): K tests the hypothesis while J tests moment validity, and the two are asymptotically independent.

Combined J–K confidence sets. The asymptotic independence of J and K motivates combining them to eliminate the spurious regions from the K confidence set (Kleibergen, 2005, Section 6). For a nominal level α , choose α_K and α_J with $\alpha_K + \alpha_J \approx \alpha$ and define the J–K confidence set as

$$CS_{JK}(\alpha) = \{\theta_0 : K(\theta_0) \leq \chi_{d_\theta}^2(1 - \alpha_K)\} \cap \{\theta_0 : J(\theta_0) \leq \chi_{d_\phi - d_\theta}^2(1 - \alpha_J)\}.$$

The J pretest prunes exactly the spurious values: wherever K is small only because the moment conditions are violated, J is large and the intersection excludes that point. In our empirical applications we use $\alpha_K = 0.04$ and $\alpha_J = 0.01$ for an overall level of $\alpha \approx 0.05$, and we report Subset- JK confidence sets for each individual elasticity and for RTS by applying the subset K inversion described above within this combined framework.

4.5.1 Identification-Robust Specification Test

As established in Section 3.5, noisy proxies do not invalidate the moment conditions: under Assumptions 2–9, identification is retained regardless of proxy error, following Doraszelski and Li (2025). Wide identification-robust confidence sets are therefore consistent with a correctly specified model in which the proxy carries limited information about productivity—they signal weak identification, not misspecification. To separate these two explanations, we need a specification test that remains valid under weak identification.

The standard Hansen J -statistic,

$$J_H(\hat{\theta}) = S(\hat{\theta}) = n \bar{\phi}_n(\hat{\theta})' \hat{V}_{\phi\phi}(\hat{\theta})^{-1} \bar{\phi}_n(\hat{\theta}),$$

is $\chi_{d_\phi - d_\theta}^2$ under correct specification and strong identification, but its limiting distribution is distorted when identification is weak because the rate of convergence of $\hat{\theta}$ depends on identification strength (Stock and Wright, 2000; Kleibergen, 2005).

Kleibergen (2005) proposes a J -statistic that remains pivotal under weak identification. Define the standardized Jacobian

$$\tilde{\mathcal{J}}_n(\theta) = \hat{V}_{\phi\phi}(\theta)^{-1/2} \hat{\mathcal{J}}_n(\theta), \quad \hat{\mathcal{J}}_n(\theta) = \frac{\partial \bar{\phi}_n(\theta)}{\partial \theta'},$$

and let

$$M_{\tilde{\mathcal{J}}(\hat{\theta})} = I_{d_\phi} - \tilde{\mathcal{J}}_n(\hat{\theta}) \left(\tilde{\mathcal{J}}_n(\hat{\theta})' \tilde{\mathcal{J}}_n(\hat{\theta}) \right)^{-1} \tilde{\mathcal{J}}_n(\hat{\theta})'$$

be the annihilator of $\tilde{\mathcal{J}}_n(\hat{\theta})$. The *Kleibergen J -statistic* is

$$KJ(\hat{\theta}) = n \bar{\phi}_n(\hat{\theta})' \hat{V}_{\phi\phi}(\hat{\theta})^{-1/2} M_{\tilde{\mathcal{J}}(\hat{\theta})} \hat{V}_{\phi\phi}(\hat{\theta})^{-1/2} \bar{\phi}_n(\hat{\theta}). \quad (30)$$

Equivalently, $KJ(\hat{\theta}) = S(\hat{\theta}) - K(\hat{\theta})$, where $K(\hat{\theta})$ is the KLM statistic of Section 4.5 evaluated at $\hat{\theta}$. Under correct specification, $KJ(\hat{\theta}) \xrightarrow{d} \chi_{d_\phi - d_\theta}^2$ irrespective of identification strength (Kleibergen, 2005). Moreover, $KJ(\hat{\theta})$ and $K(\hat{\theta})$ are asymptotically independent under H_0 : $K(\hat{\theta})$ assesses estimation precision (analogous to a first-stage F -statistic) while $KJ(\hat{\theta})$ tests overidentifying restrictions.

The practical interpretation in our setting is as follows. Failure to reject KJ combined with wide AR-GMM confidence sets is consistent with a correctly specified but weakly identified model—the moment conditions are valid, but the proxy provides limited leverage for pinning down the elasticities. Rejection of KJ points to genuine misspecification of the moment conditions irrespective of identification strength. We report KJ alongside the conventional Hansen J in our empirical applications to sharpen this distinction.

5 Monte Carlo Experiment

Our Monte Carlo design follows that of ACF closely. It builds on the analytically solvable dynamic investment model Van Biesebroeck (2003). The parameters are calibrated to reproduce key moments in the Chilean plant data analyzed by Levinsohn and Petrin (2003).

5.1 Setting

Output is produced with a Leontief production function in materials:

$$Y_{it} = \min \left\{ \alpha K_{it}^{\beta_k} L_{it}^{\beta_l} \exp(\omega_{it}), \beta_m M_{it}^* \right\} \exp(\varepsilon_{it}) \quad (31)$$

with $(\alpha, \beta_k, \beta_l, \beta_m) = (1, 0.4, 0.6, 1)$. Y_{it} denotes the output of firm i in period t ; K_{it} , L_{it} and M_{it}^* are, respectively, the levels of capital, labour and material inputs. Note that the material inputs are latent since their observations will be ridden with measurement error. ε_{it} is normal with mean zero with standard deviation 0.01. ω_{it} is assumed to follow an AR(1) with auto-regressive coefficient $\rho = 0.7$.

$$\omega_{it} = \rho \omega_{it-1} + \xi_{it} \quad (32)$$

ε_{it} is independent to the current information set and mean-zero. The variance of the normally distributed innovation ξ_{it} (σ_ξ^2) and the initial value ω_{i0} ($\sigma_{\omega_{i0}}^2$) are set such that the standard deviation of ω_{it} is equal to 0.3.

The capital stock evolves as

$$K_{it} = (1 - \delta) K_{it-1} + I_{it-1}.$$

where I_{it} denotes capital investment, and depreciation rate is set to $\delta = 0.8$. Firms choose investment to maximise the expected, discounted value of future profits subject to convex capital-adjustment costs. Labor and material inputs are selected contemporaneously with output in period t .

Because materials are used in a fixed proportion to output, the “structural value-added” regression equation (see Gandhi et al. (2020)) is

$$y_{it} = \alpha + \beta_k k_{it} + \beta_l l_{it} + \omega_{it} + \varepsilon_{it} \quad (33)$$

where lowercase letters denote the logs. The Leontief specification implies that in optimality we have that both arguments of the min-function have to be equal. This gives us the intermediate demand

function.

$$m_{it}^* = -\log(\beta_m) + \alpha + \beta_k k_{it} + \beta_l l_{it} + \omega_{it}, \quad (34)$$

where lower case letters denote the logs. We allow for independent measurement error in m_{it}^* .

$$m_{it} = m_{it}^* + \nu_{it}, \quad (35)$$

where ν_{it} is normal with zero mean and standard deviation σ_ν . We will generate data for different values of σ_ν , which adds noise to the control function. For high values of σ_ν we have weak identification, as will be shown in the results.

As ACF point out, the firm's choice on labor input L_{it} is functionally dependent on, or, in the case of measurement error, strongly correlated with, K_{it} and M_{it} . While this dependence can be accommodated in the second-stage moments, it leaves empirical identification more fragile. To strengthen identification, ACF add independent optimization error to the labor input.

$$l_{it} = l_{it}^* + \xi_{it}^l$$

where ξ_{it}^l follows a normal distribution with mean zero and a standard deviation of σ_{ξ_l} .

5.2 Simulation Results

We report one design. Measurement error in materials is added at 0%, 10%, 20%, 50%, where a 10% error raises the variance of observed materials by 10%, and the labor optimization error is held at $\sigma_{\xi_l} = 0.10$. This is the ACF design of Section 5 of Akerberg et al. (2015) with the optimization error roughly halved, which is what makes the consequences of weak identification visible at moderate measurement error. Because σ_{ξ_l} governs the independent variation in labor, lowering it moves the design toward the near-functional-dependence channel of Section 3.2; measurement error in materials then attenuates the proxy signal on top of that. The two channels of Section 3.3 therefore act together here, and the design does not separate them.

We estimate (β_l, β_k) by LP, ACF, and the one-step CUE of Section 2.3. For each replication we compute the conventional Wald test and the identification-robust S statistic of Section 4 for the null at the true parameter vector, together with the bootstrap normality test of Angelini et al. (2024).

The design is just identified, so the KLM statistic coincides with S replication by replication and we report only the latter.

Table C.1 reports the estimates. Without measurement error all three estimators are centered at the true values (0.6, 0.4) with small dispersion. As the error grows the two conventional estimators fail, and they fail differently. LP drifts monotonically: β_l rises to 0.907 at 10% and 0.972 at 50%, pressed against the upper boundary, while its capital estimates become explosive. ACF holds up through 20% error, with β_l drifting to 0.675, and then breaks down at 50%, where a nontrivial share of replications diverges. CUE remains interpretable throughout, with β_l between 0.581 and 0.599 and β_k between 0.400 and 0.404, at the cost of a standard deviation that roughly triples. Figure D.1 shows the corresponding densities: LP’s collapse to a spike at the boundary, ACF’s develop heavy tails in the same direction, and CUE’s stay centered but acquire visible skewness from 20% error onward.

Table C.2 turns to inference, and this is where the point of the paper appears. The conventional Wald test is close to nominal while identification is good, rejecting at 0.064 with no measurement error, and degrades to 0.128 at 50%—roughly two and a half times the nominal level. The identification-robust S test holds size throughout, between 0.048 and 0.062. The bootstrap normality diagnostic tracks the deterioration closely, rising from 0.065 to 0.427, which is what makes it usable as a screen: it responds to the same weakening that invalidates Wald inference.

One feature of the table deserves comment, because it bears on how the diagnostics should be read together. The Cragg–Donald test rejects underidentification in essentially every replication, at 1.000 for the first three error levels and 0.995 at 50%, even where Wald inference has clearly broken down. This is not a contradiction. The design is just identified, so T_{CD} tests a single rank deficiency against a χ^2_1 critical value, and detecting that the Jacobian has full rank is a weaker statement than establishing that it is far from rank deficient. A rank test answers whether identification holds, not whether it is strong enough for conventional inference, which is precisely the distinction drawn in Section 3.3. It is also why the empirical work of Section 6 reads T_{CD} alongside the conditional first-stage statistics and the bootstrap diagnostic rather than on its own.

6 Empirical Analysis

Our empirical analysis proceeds in two steps. We first estimate production elasticities and implied returns to scale under the ACF baseline and under CUE, using Chilean survey data (ENIA) and U.S. financial-statement data (Compustat). We then combine the estimated elasticities with input expenditure shares to construct firm-level markups.

6.1 Estimation procedure

We replicate the empirical design in Raval (2023).¹¹ Firms' log real revenue satisfies a Cobb–Douglas revenue production function in capital, labor, and materials, with productivity evolving as a linear AR(1). As in Raval (2023), we implement the ACF control-function estimator using a third-order polynomial approximation in (k, ℓ, m) for the proxy/state relationship. We then re-estimate the same specification with our one-step CUE estimator and report identification-robust inference alongside conventional large-sample approximations.

We follow Raval (2023) in variable construction, sample selection, and the ACF implementation. Output y_{it} is log real revenue and inputs $(k_{it}, \ell_{it}, m_{it})$ are log real capital, labor, and materials. In ENIA, materials are measured as intermediate consumption. In Compustat, materials are not observed and are proxied by COGS–XLR, where COGS (cost of goods sold) bundles materials, direct labor, and overhead, while XLR measures labor and related expenses with incomplete coverage. We estimate the model separately by country and industry; within each sample, the maintained specification and instrument set are held fixed across estimators, so that differences in estimates reflect the choice of estimator (ACF versus CUE) rather than changes in specification.

6.2 Specification testing

Alongside identification-robust inference we assess specification. As discussed in Section 4.5.1, wide robust confidence sets do not by themselves indicate misspecification: they are consistent with a correctly specified model in which the proxy provides limited information about productivity. To separate this from genuine violations of the moment conditions, we report the conventional Hansen J -statistic evaluated at the CUE estimate and read it against the weak-identification-robust KJ -statistic of Section 4.5.1.

¹¹Data construction details and industry definitions follow App. A of Raval (2023).

The two statistics differ precisely in their behavior under weak identification: the Hansen J is $\chi^2_{d_\phi - d_\theta}$ only when identification is strong, whereas KJ retains this limiting distribution whatever the identification strength. The asymmetry between the two directions of the test matters for how we read the results below. Both applications are just overidentified, with a single degree of freedom, and in both the Hansen J fails to reject. Rejection would have been informative: because $J \leq S$ pointwise, a rejection is evidence against the moment conditions whatever the identification strength. Non-rejection is the weaker finding, since under weak identification the Hansen J is not χ^2_1 and its size is distorted in an unknown direction. What the applications establish, then, is that the wide robust sets in Compustat cannot be attributed to a detected violation of the moment conditions, not that the model is verified there; the KJ -statistic would settle the question and we leave its computation to the revision. Rejection of KJ , by contrast, points to an underlying violation of the moment conditions—possible sources include factor-biased productivity (which would cause the output elasticities to vary with ω_{it}), within-industry heterogeneity in the production function, or errors in variable construction that contaminate the moment conditions.

6.3 Markups

We follow the production approach to markup estimation rooted in the wedge logic of Hall (1988) and implemented at the firm level by De Loecker and Warzynski (2012) and subsequent work (e.g. De Loecker et al., 2020). Let $v \in \{\ell, m\}$ be a flexible input and let θ_{it}^v denote its output elasticity implied by the estimated production function. Under static cost minimization in v conditional on state inputs, no adjustment costs, and firms taking input prices as given, the first-order condition implies

$$\mu_{it} = \frac{\theta_{it}^v}{s_{it}^v}, \quad s_{it}^v = \frac{p_{it}^v v_{it}}{P_{it} Y_{it}},$$

where s_{it}^v is the revenue share of input expenditure and μ_{it} is the price markup over marginal cost. When both labor and materials satisfy the maintained conditions, μ_{it} is invariant to the choice of v .

Raval (2023) uses this invariance as a specification check and documents systematic discrepancies between labor- and materials-based markups across datasets, emphasizing the sensitivity of production-approach markups to technology non-neutrality and measurement. De Ridder et al. (2022) further emphasize that with financial-statement data—where prices and quantities are typically unobserved—revenue-based production function estimates need not identify the level of

markups, although they can remain informative about dispersion and changes. Motivated by these insights, we report markups based on both labor and materials shares and relate their behavior to the strength of identification in the underlying elasticity estimates.

A key feature of our setting is that, in the presence of weak identification, the elasticities (and hence markups) are naturally viewed as partially identified objects. We therefore complement the standard “plug-in” markup series based on point estimates with uncertainty summaries that propagate identification-robust elasticity bounds into markups, both for time-series moments and for the cross-sectional distribution.

Let $(\underline{\beta}^v, \overline{\beta}^v)$ denote the identification-robust confidence set for the elasticity of flexible input v (obtained from the Subset- S procedure). Holding input shares fixed, this induces observation-level markup bounds

$$\underline{\mu}_{it}^v = \frac{\underline{\beta}^v}{s_{it}^v}, \quad \overline{\mu}_{it}^v = \frac{\overline{\beta}^v}{s_{it}^v},$$

where we enforce ordering of the bounds in finite samples. For each year we summarize these bounds by their cross-sectional means, $\mathbb{E}_t[\underline{\mu}_{it}^v]$ and $\mathbb{E}_t[\overline{\mu}_{it}^v]$, and overlay the corresponding mean based on the CUE point estimates, $\mathbb{E}_t[\widehat{\mu}_{it}^v]$. We report changes relative to an initial base year to focus on time variation.

To characterize uncertainty in the full cross-sectional distribution of markups, we treat each firm-year markup as interval-valued, $\mu_{it}^v \in [\underline{\mu}_{it}^v, \overline{\mu}_{it}^v]$, and construct two complementary distributional summaries.

We summarize these interval-valued markups with the *exact interval-mixture density* implied by the observation-level bounds. Specifically, each firm-year contributes a uniform density on its interval $[\underline{\mu}_{it}^v, \overline{\mu}_{it}^v]$, and the cross-sectional density is the equal-weight mixture of these observation-specific uniforms. This yields the density

$$f_{\mu^v}(x) = \frac{1}{N_t} \sum_{i=1}^{N_t} \frac{\mathbf{1}\{x \in [\underline{\mu}_{it}^v, \overline{\mu}_{it}^v]\}}{\overline{\mu}_{it}^v - \underline{\mu}_{it}^v},$$

defined wherever the interval width is positive. We overlay this exact interval-mixture density with the kernel density of the point-estimate markups $\widehat{\mu}_{it}^v$ computed from CUE elasticities. The comparison illustrates how much of the cross-sectional distribution is driven by identification uncertainty in the elasticities, versus dispersion in the plug-in point estimates themselves.

6.4 Data

We use two datasets that differ in measurement quality. The Chilean data are from ENIA and are plant-level, survey-based observations for *Manufacture of food products* (ISIC 311) over 1980–1996, comprising approximately 16,000 plant–year observations. Materials are recorded as intermediate consumption and are cleanly separated from labor compensation, yielding a relatively precise proxy for productivity.

The U.S. data are from Compustat and are firm-level accounting data for *Food Manufacturing* (NAICS 31) over 1970–2005, comprising approximately 1,000 firm–year observations. Relative to ENIA, *all* key inputs are measured with greater noise: materials are constructed from accounting aggregates (COGS–XLR), capital is a book-value stock subject to accounting conventions, and labor costs are imputed from aggregate wages. This lowers the overall signal-to-noise ratio across the input vector rather than compromising any single proxy mapping, and weak-identification concerns are correspondingly more salient in the U.S. sample. We therefore attribute identification difficulties in Compustat to this general deterioration in data quality rather than to a specific proxy invertibility failure in the LP sense (see Remark 1).

6.5 Chile (ENIA)

Table C.3 reports results for the Chilean ENIA sample (ISIC 311). The estimates are stable across estimators. Under ACF, elasticities are $\hat{\beta}_k = 0.040$, $\hat{\beta}_\ell = 0.123$, and $\hat{\beta}_m = 0.905$, implying mildly increasing returns to scale (RTS= 1.067). The CUE estimates are similar— $\hat{\beta}_k = 0.014$, $\hat{\beta}_\ell = 0.110$, and $\hat{\beta}_m = 0.948$ —with RTS= 1.072.

The Cragg–Donald minimum-eigenvalue statistic is $T_{CD} = 336.1$ (χ^2_2 critical value at 95%: 5.99, $p < 0.001$), decisively rejecting the null of underidentification and confirming that the Jacobian has full rank. As a complementary diagnostic of conditional instrument relevance for the labor and materials quasi-differences $\ell_{it} - \rho\ell_{it-1}$ and $m_{it} - \rho m_{it-1}$, the conditional first-stage F -statistics of Section 4.2 are $F_\ell = 481$ and $F_m = 529$; we interpret these comparatively rather than against linear-IV critical values. This is reflected in the confidence sets: the Subset- S intervals for β_ℓ and β_m are only modestly wider than the corresponding Wald intervals, and the J–K intervals are of comparable width. Specifically, the Subset- S 95% sets are $[0.004, 0.025]$ for β_k , $[0.083, 0.143]$ for β_ℓ , $[0.923, 0.968]$ for β_m , and $[1.055, 1.091]$ for RTS; the corresponding J–K sets are $[0.006, 0.034]$, $[0.058, 0.118]$, $[0.934, 0.978]$, and $[1.030, 1.090]$. The Hansen J -test does not reject at CUE ($J = 0.258$, $\text{df} = 1$,

$p = 0.612$), and here identification is strong enough that its χ^2_1 approximation can be relied on, so the non-rejection is genuine evidence of correct specification.

Bootstrap diagnostics reinforce this benign picture. The Shapiro–Wilk test fails to reject normality for all three elasticities ($p = 0.698$ for β_k , $p = 0.556$ for β_ℓ , and $p = 0.303$ for β_m), the corresponding Jarque–Bera tests also fail to reject ($p = 0.940, 0.771$, and 0.469), and the Doornik–Hansen joint test likewise does not reject ($p = 0.943$). Taken together, these diagnostics are consistent with approximately standard large-sample behavior in the Chilean sample: the Jacobian is far from rank-deficient, the robust confidence sets are well-bounded, and the bootstrap draws do not exhibit the pronounced non-Gaussian features seen in the U.S. data.

Figure D.2 plots the evolution of average markups in Chile (ENIA, ISIC 311), separately for labor- and materials-based markups. The solid line reports the annual mean of the plug-in markups constructed from the CUE point estimates. The shaded band reports the corresponding year-by-year bounds obtained by propagating the identification-robust elasticity intervals into observation-level markup intervals and averaging within year.¹² The bounds are relatively tight throughout the sample and track the point series closely, indicating that the time variation in average markups is informative and not driven by identification uncertainty in the underlying elasticities.

Figure D.3 compares two estimates of the cross-sectional distribution of markups in a representative year (1996). The solid curve reports the exact interval-mixture density obtained by averaging the observation-specific uniform densities on $[\underline{\mu}_{it}^v, \bar{\mu}_{it}^v]$, while the dashed curve reports the kernel density of plug-in CUE markups $\hat{\mu}_{it}^v$. The two densities are close, and the interval-mixture distribution is only slightly more dispersed, suggesting that identification uncertainty has limited distributional consequences in this sample. Relative to the U.S. case, the markup inference is well-behaved, consistent with a strong proxy signal in ENIA where materials are measured directly as intermediate consumption.

6.6 U.S. (Compustat)

Table C.4 reports results for the U.S. Compustat sample (NAICS 31). The estimates are considerably less stable than in the Chilean survey data. Under ACF, elasticities are $\hat{\beta}_k = 0.329$, $\hat{\beta}_\ell = 0.132$, and $\hat{\beta}_m = 0.561$, implying approximately constant returns to scale (RTS= 1.022). In contrast, the CUE estimates differ sharply: $\hat{\beta}_k = 1.696$, $\hat{\beta}_\ell = -0.992$, and $\hat{\beta}_m = -0.537$, implying RTS= 0.167.

¹²In each year we compute $\underline{\mu}_{it}^v = \underline{\beta}^v / s_{it}^v$ and $\bar{\mu}_{it}^v = \bar{\beta}^v / s_{it}^v$ and then take cross-sectional means. The figure reports changes relative to the first sample year.

This large divergence between ACF and CUE—with sign reversals for labor and materials and economically implausible point estimates—is characteristic of weak identification: when the GMM criterion is nearly flat, small changes in weighting produce large swings in the minimizer.

Identification-robust inference directly confirms this picture. The Cragg–Donald minimum-eigenvalue statistic fails to reject underidentification ($T_{CD} = 0.093$, $df = 2$, $p = 0.955$), indicating that the Jacobian is near-rank-deficient and the moment conditions contain very limited information about the structural parameters. The conditional first-stage F -statistics of Section 4.2 for the labor and materials quasi-differences are essentially zero ($F_\ell = 0.015$ and $F_m = 0.091$), in stark contrast to ENIA, indicating that after partialling out the other quasi-differenced inputs the instruments have virtually no remaining relevance for either static input. The Subset- S confidence sets are correspondingly wide: $[-0.100, 2.500]$ for β_k , $[-4.790, 1.810]$ for β_ℓ , $[-2.540, 0.660]$ for β_m , and $[-0.430, 0.770]$ for RTS. The Wald intervals are enormous and uninformative. Bootstrap normality is strongly rejected for all elasticities (Shapiro–Wilk $p < 0.001$ for each; Doornik–Hansen joint $p \approx 0$), consistent with the highly non-Gaussian sampling behavior predicted by Proposition 8 under weak identification.

The Subset- JK confidence sets remain wide and are left-unbounded for labor and materials (and both-sided unbounded for RTS): $[0.200, 2.700]$ for β_k , $[<-5.990, 1.110]$ for β_ℓ , $[<-5.540, 1.260]$ for β_m , and $[<-4.830, >5.170]$ for RTS. In this weakly identified setting these intervals should be read as confirming that the data provide little information about the elasticities, rather than as a failure of the test: several upper bounds are informative while open boundaries are driven by the grid.

On specification, the conventional Hansen J -test does not reject at the CUE estimate ($J = 0.905$, $df = 1$, $p = 0.341$). Because identification is weak here, the Hansen J need not have the correct size, so non-rejection is suggestive rather than conclusive: it establishes that no violation of the moment conditions is detected, not that none is present. Read with that caveat, the evidence points away from model misspecification and toward limited identifying variation in the input quasi-differences once accounting construction and high persistence are taken into account: the moment conditions appear valid, but they have limited leverage for pinning down the elasticities. This pattern is consistent with the near-functional-dependence channel in Section 3 and with the weak-ID Monte Carlo evidence of Section 5.

Figure D.4 reports the U.S. (Compustat, NAICS 31) time series of mean markups, constructed analogously to the Chilean case. The solid line is the annual mean of plug-in CUE markups $\mathbb{E}_t[\hat{\mu}_{it}^v]$,

while the shaded band plots the corresponding mean bounds $\mathbb{E}_t[\underline{\mu}_{it}^v]$ and $\mathbb{E}_t[\overline{\mu}_{it}^v]$ induced by the identification-robust elasticity intervals. In stark contrast to ENIA, the bounds are extremely wide throughout the sample and often extend far into economically implausible regions, so that changes in average markups are only weakly informative once elasticity uncertainty is propagated. This pattern is consistent with the weak-identification features already evident in the elasticity estimates and their robust confidence sets.

Figure D.5 illustrates the same issue for the cross-sectional distribution in 1996. The dashed kernel density based on plug-in CUE markups is tightly concentrated, reflecting the cross-sectional dispersion in the point estimates. By contrast, the solid exact interval-mixture density based on interval markups $\mu_{it}^v \in [\underline{\mu}_{it}^v, \overline{\mu}_{it}^v]$ is extremely diffuse and nearly flat over a wide range. This gap indicates that, in the U.S. sample, a substantial part of what appears as a precise distribution under plug-in markups is not supported once identification-robust elasticity uncertainty is accounted for: the data deliver limited information about the level and dispersion of markups in this setting, in line with concerns raised by De Ridder et al. (2022) for financial-statement data.

7 Conclusion

This paper has studied control-function estimation of production functions under weak identification. The elasticities can be point identified while the moment conditions carry very little information about them, and the two cases call for different inference.

What makes the distinction operational is the orthogonalized Jacobian of Section 3: the second-stage Jacobian once the productivity dynamics have been projected out. Its rank is the local identification condition, which extends the rank condition of Robinson (1988) to this semiparametric setting, and its scaled cross-product measures identification strength. Two features of the model push it toward rank deficiency. Because static inputs respond to productivity, quasi-differenced inputs are close to collinear once lagged productivity is conditioned on; and noise in the proxy, whether from optimization frictions, unobserved variation in input prices, or inputs reconstructed from accounting aggregates, attenuates the identifying signal. Either can be made severe without violating any assumption under which the estimator is consistent, and as it becomes severe the continuously updated estimator ceases to be asymptotically normal and Wald intervals undercover.

Identification strength should therefore be reported, and inference should not presume it. We

adapt three tools to this setting: a bootstrap diagnostic for the normality of the estimator’s sampling distribution, identification-robust confidence sets together with their subset versions, and a robust test of the overidentifying restrictions, which the conventional Hansen test cannot provide once identification is weak. The simulation evidence shows the pattern they are meant to catch. As the identifying signal weakens, Wald inference deteriorates while the robust tests keep size, and a rank test goes on rejecting underidentification throughout—full rank being a much weaker property than distance from rank deficiency.

The two applications differ as the theory predicts. In Chilean plant data, where materials are measured as intermediate consumption, estimates are stable across estimators, the diagnostics are favorable, and the robust intervals imply informative markup bounds. In Compustat data for the same industry, where inputs must be reconstructed from accounting aggregates, identification is weak by every diagnostic we report, the robust set for the labor elasticity is wide, and the markup bounds are uninformative. The Hansen test rejects in neither sample, and the point estimates and standard errors ordinarily reported give no indication of the difference.

For applied work in the OP, LP, and ACF tradition we recommend reporting identification diagnostics alongside the usual standard errors, using robust confidence sets when the diagnostics call for it, and propagating elasticity uncertainty into markups rather than plugging in point estimates. Two limitations suggest further work. The robust test of the overidentifying restrictions is developed here but not computed in the applications. And the robust procedures we use are not tailored to the near-rank-deficiency that arises in this model; tests that exploit its structure should have better power.

References

- Akerberg, D. A., Caves, K., and Frazer, G. (2015). Identification properties of recent production function estimators. *Econometrica*, 83(6):2411–2451.
- Akerberg, D. A., Lanier Benkard, C., Berry, S., and Pakes, A. (2007). Chapter 63 econometric tools for analyzing market outcomes. volume 6 of *Handbook of Econometrics*, pages 4171–4276. Elsevier.
- Anderson, T. W. and Rubin, H. (1949). Estimation of the parameters of a single equation in a complete set of stochastic equations. *The Annals of Mathematical Statistics*, 21:570–582.
- Andrews, D. W. and Stock, J. H. (2005). Inference with weak instruments. Technical Working Paper 313, National Bureau of Economic Research, Cambridge, MA.
- Andrews, D. W. K. (1994). Empirical process methods in econometrics. In Engle, R. F. and McFadden, D. L., editors, *Handbook of Econometrics*, volume 4, chapter 37, pages 2247–2294. Elsevier.
- Andrews, I. and Mikusheva, A. (2015). Maximum likelihood inference in weakly identified dynamic stochastic general equilibrium models. *Quantitative Economics*, 6(1):123–152.

- Angelini, G., Cavaliere, G., and Fanelli, L. (2024). An identification and testing strategy for proxy-svars with weak proxies. *Journal of Econometrics*, 238(2):105604.
- Canova, F. and Sala, L. (2009). Back to square one: Identification issues in DSGE models. *Journal of Monetary Economics*, 56(4):431–449.
- Cragg, J. G. and Donald, S. G. (1993). Testing identifiability and specification in instrumental variable models. *Econometric Theory*, 9(2):222–240.
- De Loecker, J., Eeckhout, J., and Unger, G. (2020). The rise of market power and the macroeconomic implications. *The Quarterly Journal of Economics*, 135(2).
- De Loecker, J. and Warzynski, F. (2012). Markups and firm-level export status. *American Economic Review*, 102(6):2437–2471.
- De Ridder, M., Grassi, B., and Morzenti, G. (2022). Dp17532 the hitchhiker’s guide to markup estimation: Assessing estimates from financial data. CEPR Discussion Paper 17532, CEPR, Paris and London.
- Doraszelski, U. and Li, L. (2025). Production function estimation without invertibility: Imperfectly competitive environments and demand shocks. Working paper, 2025.
- Dufour, J.-M. (1997). Some impossibility theorems in econometrics with applications to structural and dynamic models. *Econometrica*, 65(6):1365–1387.
- Gandhi, A., Navarro, S., and Rivers, D. A. (2020). On the identification of gross output production functions. *Journal of Political Economy*, 128(8). Electronically published.
- Hall, R. (1988). The relation between price and marginal cost in u.s. industry. *Journal of Political Economy*, 96(5):921–47.
- Hansen, L. P. (1982). Large sample properties of generalized method of moments estimators. *Econometrica*, 50(4):1029–1054.
- Kleibergen, F. (2002). Pivotal statistics for testing structural parameters in instrumental variables regression. *Econometrica*, 70(5):1781–1803.
- Kleibergen, F. (2005). Testing parameters in gmm without assuming that they are identified. *Econometrica*, 73:1103–1123.
- Kleibergen, F. and Mavroeidis, S. (2009a). Inference on subsets of parameters in GMM without assuming identification. Working paper (updated version), Brown University.
- Kleibergen, F. and Mavroeidis, S. (2009b). Weak instrument robust tests in gmm and the new keynesian phillips curve. *Journal of Business & Economic Statistics*, 27(3):293–311.
- Levinsohn, J. A. and Petrin, A. (2003). Estimating production functions using inputs to control for unobservables. *Review of Economic Studies*, 70(2):317–340.
- Mavroeidis, S. (2005). Identification issues in forward-looking models estimated by GMM, with an application to the phillips curve. *Journal of Money, Credit and Banking*, 37(3):421–448.
- Moreira, M. J. (2003). A conditional likelihood ratio test for structural models. *Econometrica*, 71(4):1027–1048.
- Newey, W. K. and McFadden, D. (1994). Large sample estimation and hypothesis testing. In Engle, R. F. and McFadden, D., editors, *Handbook of Econometrics*, volume 4, chapter 36, pages 2111–2245. Elsevier.
- Newey, W. K. and West, K. D. (1987). Hypothesis testing with efficient method of moments estimation. *International Economic Review*, 28(3):777–787.
- Olley, G. S. and Pakes, A. (1996). The dynamics of productivity in the telecommunications equipment industry. *Econometrica*, 64(6):1263–1297.
- Raval, D. (2023). Testing the Production Approach to Markup Estimation. *The Review of Economic Studies*, 90(5):2592–2611.
- Robinson, P. M. (1988). Root-N-consistent semiparametric regression. *Econometrica*, 56(4):931–954.
- Sanderson, E. and Windmeijer, F. (2016). A weak instrument f-test in linear iv models with multiple endogenous variables. *Journal of Econometrics*, 190(2):212–221.
- Staiger, D. and Stock, J. H. (1997). Instrumental variables regression with weak instruments. *Econometrica*, 65(3):557–586.

- Stock, J. H. and Wright, J. H. (2000). Gmm with weak identification. *Econometrica*, 68(5):1055–1096.
- Stock, J. H., Wright, J. H., and Yogo, M. (2002). A survey of weak instruments and weak identification in generalized method of moments. *Journal of Business & Economic Statistics*, 20(4):518–529.
- Van Biesebroeck, J. (2003). Productivity dynamics with technology choice: An application to automobile assembly. *Review of Economic Studies*, 70(1):167–198.
- Wang, J. and Zivot, E. (1998). Inference on structural parameters in instrumental variables regression with weak instruments. *Econometrica*, 66(6):1389–1404.
- Wooldridge, J. (2009). On estimating firm-level production variables to control for unobservables. *Economics Letters*, 104:112–114.
- Wright, J. H. (2003). Detecting lack of identification in GMM. *Econometric Theory*, 19(3):322–330.
- Yogo, M. (2004). Estimating the elasticity of intertemporal substitution when instruments are weak. *Review of Economics and Statistics*, 86(3):797–810.

Appendices

A Proofs of Section 3

Lemma 10. Suppose $m_n(\vartheta_w, \vartheta_{s0})$ is continuously differentiable in ϑ_w on a compact convex set Θ_w , $m_n(\vartheta_{w0}, \vartheta_{s0}) = 0$, and $\partial m_n(\vartheta_w, \vartheta_{s0}) / \partial \vartheta'_w = M_{w,n}(\vartheta_w, \vartheta_{s0})$. If $M_{w,n}(\vartheta_w, \vartheta_{s0}) = n^{-1/2}C(\vartheta_w, \vartheta_{s0}) + o(n^{-1/2})$ uniformly for $\vartheta_w \in \Theta_w$ and for a bounded function $C(\vartheta_w, \vartheta_{s0})$ uniformly for $\vartheta_w \in \Theta_w$, then

$$m_n(\vartheta_w, \vartheta_{s0}) = n^{-1/2}m_1(\vartheta_w) + o(n^{-1/2}) \quad \text{for all } \vartheta_w \in \Theta_w,$$

where

$$m_1(\vartheta_w) := \int_0^1 C(\vartheta_{w0} + t(\vartheta_w - \vartheta_{w0}), \vartheta_{s0})(\vartheta_w - \vartheta_{w0}) dt.$$

In particular, $m_n(\vartheta_w, \vartheta_{s0}) = n^{-1/2}m_1(\vartheta_w) + o(n^{-1/2})$ uniformly on Θ_w .

Proof. Fix $\vartheta_w \in \Theta_w$ and define the line segment $\vartheta_{w,t} := \vartheta_{w0} + t(\vartheta_w - \vartheta_{w0}) \in \Theta_w$ for $t \in [0, 1]$. By the fundamental theorem of calculus and the assumed interchange of differentiation and expectation,

$$\begin{aligned} m_n(\vartheta_w, \vartheta_{s0}) - m_n(\vartheta_{w0}, \vartheta_{s0}) &= \int_0^1 \frac{d}{dt} m_n(\vartheta_{w,t}, \vartheta_{s0}) dt \\ &= \int_0^1 \frac{\partial m_n(\vartheta_{w,t}, \vartheta_{s0})}{\partial \vartheta'_w} (\vartheta_w - \vartheta_{w0}) dt \\ &= \int_0^1 M_{w,n}(\vartheta_{w,t}, \vartheta_{s0})(\vartheta_w - \vartheta_{w0}) dt. \end{aligned}$$

Using $m_n(\vartheta_{w0}, \vartheta_{s0}) = 0$ and $M_{w,n}(\vartheta_{w,t}, \vartheta_{s0}) = n^{-1/2}C(\vartheta_{w,t}, \vartheta_{s0}) + o(n^{-1/2})$ yields

$$m_n(\vartheta_w, \vartheta_{s0}) = n^{-1/2} \int_0^1 C(\vartheta_{w0} + t(\vartheta_w - \vartheta_{w0}), \vartheta_{s0})(\vartheta_w - \vartheta_{w0}) dt + o(n^{-1/2}) = n^{-1/2}m_1(\vartheta_w) + o(n^{-1/2}),$$

which proves the claim. $\square$

Proof of Proposition 7. We have that,

$$\begin{aligned} m_n(\vartheta_w, \vartheta_s) &= m_n(\vartheta_w, \vartheta_{s0}) + R(\vartheta_w)(\vartheta_s - \vartheta_{s0}) \\ &\quad + \underbrace{\left(M_{w,n}(\vartheta_w, \vartheta_s) - M_{w,n}(\vartheta_w, \vartheta_{s0}) \right)}_{O(n^{-1/2})} (\vartheta_w - \vartheta_{w0}) + r_n(\vartheta_w, \vartheta_s) - r_n(\vartheta_w, \vartheta_{s0}), \end{aligned}$$

The $O(n^{-1/2})$ -rate comes from Assumption 4 i), compactness of Θ_w , Assumption 4, (iv): the original remainder r_n is uniformly $o(n^{-1/2})$. By Lemma 10, we can write

$$m_n(\vartheta_w, \vartheta_s) = n^{-1/2}m_1(\vartheta_w) + R(\vartheta_w)(\vartheta_s - \vartheta_{s0}) + O(n^{-1/2})$$

Consistency. By a uniform LLN and $W_n \xrightarrow{p} W_0$,

$$\frac{1}{n} Q_n(\vartheta_w, \vartheta_s) = Q_{0,n}(\vartheta_w, \vartheta_s) + o_p(1)$$

where, $Q_{0,n}(\vartheta_w, \vartheta_s) = m_n(\vartheta_w, \vartheta_s)' W_0 m_n(\vartheta_w, \vartheta_s)$. By Lemma 10, the decomposition above, and

boundedness of $R(\vartheta_w)$ on Θ_w ,

$$Q_{0,n} = (\vartheta_s - \vartheta_{s0})' R(\vartheta_w)' W_0 R(\vartheta_w) (\vartheta_s - \vartheta_{s0}) + O(n^{-1/2}),$$

uniformly on $\Theta_w \times \Theta_s$.

Since W_0 is positive definite and $R(\vartheta_w)$ has full column rank uniformly on Θ_w , there exists $c > 0$ such that

$$(\vartheta_s - \vartheta_{s0})' R(\vartheta_w)' W_0 R(\vartheta_w) (\vartheta_s - \vartheta_{s0}) \geq c \|\vartheta_s - \vartheta_{s0}\|^2 \quad \forall \vartheta_w \in \Theta_w.$$

Fix any $\epsilon > 0$. Then for all $\vartheta_w \in \Theta_w$ and all ϑ_s such that $\|\vartheta_s - \vartheta_{s0}\| \geq \epsilon$,

$$Q_{0,n}(\vartheta_w, \vartheta_s) \geq c\epsilon^2 + O(n^{-1/2}).$$

Hence, for n sufficiently large,

$$\inf_{\vartheta_w \in \Theta_w, \|\vartheta_s - \vartheta_{s0}\| \geq \epsilon} Q_{0,n}(\vartheta_w, \vartheta_s) \geq \frac{c\epsilon^2}{2}.$$

Note that $Q_{0,n}(\vartheta_{w0}, \vartheta_{s0}) = 0$, so the population criterion is uniformly separated away from zero outside any ϵ -neighborhood of ϑ_{s0} . And since $\frac{1}{n}Q_n$ is uniformly approximated by $Q_{0,n}$, standard extremum-estimator arguments imply $\hat{\vartheta}_s \xrightarrow{P} \vartheta_{s0}$.

$\sqrt{n}$ -consistency. By the previous step, $\hat{\vartheta}_s \xrightarrow{P} \vartheta_{s0}$, and by Assumption 6, $\lambda_{\min}(W_n) \xrightarrow{P} \lambda_{\min}(W_0) > 0$. Moreover, under Assumption 5, $\sup_{\vartheta \in \Theta_w \times \Theta_s} \|\psi_n(\vartheta)\| = O_p(1)$.

Since $\hat{\vartheta}$ minimizes $Q_n(\vartheta)$, we have $Q_n(\hat{\vartheta}) \leq Q_n(\vartheta_0)$, hence

$$\lambda_{\min}(W_n) \|\psi_n(\hat{\vartheta}) + \sqrt{n} m_n(\hat{\vartheta})\|^2 \leq Q_n(\hat{\vartheta}) \leq Q_n(\vartheta_0) = \psi_n(\vartheta_0)' W_n \psi_n(\vartheta_0) = O_p(1).$$

Therefore, $\|\psi_n(\hat{\vartheta}) + \sqrt{n} m_n(\hat{\vartheta})\| = O_p(1)$, and hence $\|\sqrt{n} m_n(\hat{\vartheta})\| = O_p(1)$.

Using the same expansion as above of $m_n(\vartheta_w, \vartheta_s)$ in ϑ_s ,

$$\sqrt{n} m_n(\hat{\vartheta}_w, \hat{\vartheta}_s) = \sqrt{n} m_n(\hat{\vartheta}_w, \vartheta_{s0}) + R(\hat{\vartheta}_w) \sqrt{n}(\hat{\vartheta}_s - \vartheta_{s0}) + O(1).$$

By Lemma 10, $\sup_{\vartheta_w \in \Theta_w} \|\sqrt{n} m_n(\vartheta_w, \vartheta_{s0})\| = O(1)$, so $\|\sqrt{n} m_n(\hat{\vartheta}_w, \vartheta_{s0})\| = O(1)$. Combining with $\|\sqrt{n} m_n(\hat{\vartheta})\| = O_p(1)$ gives

$$\|R(\hat{\vartheta}_w) \sqrt{n}(\hat{\vartheta}_s - \vartheta_{s0})\| = O_p(1).$$

Since $[C(\vartheta_w, \vartheta_s) R(\vartheta_w)]$ has full column rank $d_w + d_s$ uniformly on $\Theta_w \times \Theta_s$ by Assumption 4(iii), $R(\vartheta_w)$ itself has full column rank d_s uniformly on Θ_w (if $R(\vartheta_w)x = 0$ then $[C R] \begin{pmatrix} 0 \\ x \end{pmatrix} = 0$, contradicting full column rank). Since R is continuous on the compact set Θ_w , there exists $c_R > 0$ such that $\|R(\vartheta_w)x\| \geq c_R \|x\|$ for all $x \in \mathbb{R}^{d_s}$ and all $\vartheta_w \in \Theta_w$. Applying this with $\vartheta_w = \hat{\vartheta}_w$ yields $\sqrt{n}(\hat{\vartheta}_s - \vartheta_{s0}) = O_p(1)$. $\square$

Proof of Proposition 8. Let $\tilde{\vartheta}_s = \vartheta_{s0} + \lambda/\sqrt{n}$ for fixed $\lambda \in \mathbb{R}^{d_s}$. Using Lemma 10 we can write

$$\sqrt{n} m_n(\vartheta_w, \tilde{\vartheta}_s) = m_1(\vartheta_w) + R(\vartheta_w)\lambda + O(1),$$

uniformly in $\vartheta_w \in \Theta_w$. Moreover, since Θ_w is compact and $m_1(\vartheta_w)$ and $R(\vartheta_w)$ are bounded on Θ_w ,

$$\sup_{\vartheta_w \in \Theta_w} \|\sqrt{n} m_n(\vartheta_w, \tilde{\vartheta}_s)\| = O(1).$$

By Assumption 5, $\psi_n \Rightarrow \psi_0$, where ψ_0 has almost surely bounded sample paths. Since $\tilde{\vartheta}_s - \vartheta_{s0} \rightarrow$

0, asymptotic stochastic equicontinuity of ψ_n implies

$$\sup_{\vartheta_w \in \Theta_w} \|\psi_n(\vartheta_w, \tilde{\vartheta}_s) - \psi_n(\vartheta_w, \vartheta_{s0})\| \xrightarrow{p} 0.$$

Hence

$$\sup_{\vartheta_w \in \Theta_w} \|\psi_n(\vartheta_w, \tilde{\vartheta}_s)\| = O_p(1).$$

Therefore, using

$$\sqrt{n} \bar{\phi}_n(\vartheta_w, \tilde{\vartheta}_s) = \psi_n(\vartheta_w, \tilde{\vartheta}_s) + \sqrt{n} m_n(\vartheta_w, \tilde{\vartheta}_s),$$

we obtain

$$\sup_{\vartheta_w \in \Theta_w} \|\sqrt{n} \bar{\phi}_n(\vartheta_w, \tilde{\vartheta}_s)\| = O_p(1).$$

Since $W_n \xrightarrow{p} W_0$, we have $\|W_n\| = O_p(1)$, and therefore

$$\begin{aligned} \sup_{\vartheta_w \in \Theta_w} Q_n(\vartheta_w, \tilde{\vartheta}_s) &= \sup_{\vartheta_w \in \Theta_w} \left[\left(\sqrt{n} \bar{\phi}_n(\vartheta_w, \tilde{\vartheta}_s) \right)' W_n \left(\sqrt{n} \bar{\phi}_n(\vartheta_w, \tilde{\vartheta}_s) \right) \right] \\ &\leq \|W_n\| \sup_{\vartheta_w \in \Theta_w} \|\sqrt{n} \bar{\phi}_n(\vartheta_w, \tilde{\vartheta}_s)\|^2 = O_p(1). \end{aligned}$$

□

B Monte Carlo Design

We simulate $n = 1000$ firms observed for $T = 10$ consecutive periods. Unless otherwise noted, parameter values are taken from ACF. The parameters are calibrated so that roughly 95% of the cross-sectional variation in capital is between firms and a fixed-effects regression of k_{it} on l_{it} delivers $R^2 \approx 0.50$.

B.1 Choice of Labour and Material Inputs

Conditional on its information set, the firm solves

$$\max_{L_{it}, M_{it}} \alpha K_{it}^{\beta_k} L_{it}^{\beta_l} \exp(\omega_{it}) \quad \text{s.t.} \quad M_{it} \geq \beta_m^{-1} \alpha K_{it}^{\beta_k} L_{it}^{\beta_l} \exp(\omega_{it}),$$

yielding in optimality

$$\begin{aligned} L_{it}^* &= \left(\frac{\alpha \beta_l}{W_{it}} \right)^{\frac{1}{1-\beta_l}} K_{it}^{\frac{\beta_k}{1-\beta_l}} \exp \left(\frac{\omega_{it}}{1-\beta_l} \right) \\ M_{it}^* &= \beta_m^{-1} \alpha K_{it}^{\beta_k} L_{it}^{*\beta_l} \exp(\omega_{it}) \end{aligned}$$

Following ACF, we contaminate L_{it}^* with iid optimisation noise to break the functional dependence issue.

$$L_{it} = L_{it}^* \exp(\xi_{it}^l)$$

where ξ_{it}^l follows a normal distribution with mean zero and a standard deviation of 0.1. ACF chooses a standard deviation of the optimisation error of 0.37. We deliberately adopt a smaller value. A larger variance indeed strengthens identification by injecting more cross-sectional variation into the labour input and thereby mitigating—but not eliminating—the weak-instrument problem. For expositional

clarity, however, a modest reduction in the variance still breaks the functional dependence at the heart of the ACF critique while keeping the simulated distributions readily interpretable.

B.2 Investment Choice and Capital Accumulation

Capital is a dynamic input. Investment is chosen at $t-1$ and the capital stock evolves as

$$K_{i,t+1} = (1 - \delta)K_{it} + I_{it}, \quad c_i(I_{it}) = \frac{\phi_i}{2} I_{it}^2,$$

where $\phi_i > 0$ is firm-specific; we draw ϕ_i^{-1} lognormally with cross-sectional standard deviation 0.6. This implies a strong heterogeneity, with a substantial mass of firms having very high adjustment costs. Firms maximize the expected present value of profits net of adjustment costs. With the static labor choice contaminated by multiplicative optimization error,

$$L_{it} = L_{it}^* \exp(\xi_{it}^l), \quad \xi_{it}^l \sim \mathcal{N}(0, \sigma_{\xi_l}^2) \text{ i.i.d.},$$

the (expected) marginal value of one more unit of installed capital next period is

$$\text{MVPK}_{i,t+1} = \left(\frac{\beta_k}{1 - \beta_l} \right) \alpha^{\frac{1}{1-\beta_l}} W_{i,t+1}^{-\frac{\beta_l}{1-\beta_l}} \exp\left(\frac{\omega_{i,t+1}}{1 - \beta_l}\right) \left[\beta_l^{\frac{\beta_l}{1-\beta_l}} e^{\frac{1}{2}\beta_l^2 \sigma_{\xi_l}^2} - \beta_l^{\frac{1}{1-\beta_l}} e^{\frac{1}{2}\sigma_{\xi_l}^2} \right].$$

The term in square brackets is due to the optimization-error in labor. The Euler equation with quadratic adjustment costs yields a closed-form policy:

$$I_{it} = \frac{\beta}{\phi_i} \sum_{\tau=0}^{\infty} (\beta(1 - \delta))^{\tau} \mathbb{E}_t[\text{MVPK}_{i,t+1+\tau}].$$

We assume $\ln W_{it}$ follows an independent covariance-stationary AR(1):

$$\ln W_{t+h} = \rho_W^h \ln W_t + \zeta_{t+h},$$

with $\zeta_{t+h} \sim \mathcal{N}(0, \sigma_{\zeta}^2 \sum_{j=0}^{h-1} \rho_W^{2j})$, independent of current information. Then conditional log-normality implies, for $h = \tau + 1$,

$$\mathbb{E}_t \left[e^{\frac{\omega_{t+h}}{1-\beta_l}} W_{t+h}^{-\frac{\beta_l}{1-\beta_l}} \right] = \exp \left(\frac{\rho^h}{1 - \beta_l} \omega_t - \frac{\rho_W^h \beta_l}{1 - \beta_l} \ln W_t + \frac{1}{2} \frac{V_{\omega, h-1}}{(1 - \beta_l)^2} + \frac{1}{2} \frac{\beta_l^2 V_{W, h-1}}{(1 - \beta_l)^2} \right),$$

where the forecast-error variances have closed forms (for $|\rho|, |\rho_W| < 1$):

$$V_{\omega, h-1} = \sigma_{\xi}^2 \sum_{j=0}^{h-1} \rho^{2j} = \sigma_{\xi}^2 \frac{1 - \rho^{2h}}{1 - \rho^2}, \quad V_{W, h-1} = \sigma_{\zeta}^2 \sum_{j=0}^{h-1} \rho_W^{2j} = \sigma_{\zeta}^2 \frac{1 - \rho_W^{2h}}{1 - \rho_W^2}.$$

Substituting the last display into the Euler sum gives a fully explicit expression for I_{it} .

$$I_{it} = \frac{\beta}{\phi_i} \frac{\beta_k}{1 - \beta_{\ell}} \alpha^{\frac{1}{1-\beta_{\ell}}} B_{\xi} \sum_{\tau=0}^{\infty} [\beta(1 - \delta)]^{\tau} \exp \left(\frac{\rho^{\tau+1}}{1 - \beta_{\ell}} \omega_{it} - \frac{\rho_W^{\tau+1} \beta_{\ell}}{1 - \beta_{\ell}} \ln W_{it} + \frac{1}{2} V_{\tau} \right),$$

with

$$B_{\xi} = \beta_{\ell}^{\frac{\beta_{\ell}}{1-\beta_{\ell}}} e^{\frac{1}{2}\beta_{\ell}^2 \sigma_{\xi_{\ell}}^2} - \beta_{\ell}^{\frac{1}{1-\beta_{\ell}}} e^{\frac{1}{2}\sigma_{\xi_{\ell}}^2}, \quad V_{\tau} = \frac{V_{\omega, \tau}}{(1 - \beta_{\ell})^2} + \frac{\beta_{\ell}^2 V_{W, \tau}}{(1 - \beta_{\ell})^2}$$

B.3 Steady-State Initialisation

All firms start at $K_{i0} = 0$. We simulate the model forward until the cross-section of capital stocks converges to the stationary distribution implied by the policy rule; 10 consecutive periods from that steady state are retained for estimation.

B.4 Estimation

We apply the ACF and the LP estimators described in ACF Appendix A. The CUE is described in Section 4. We use moment vector

$$f_{it}(\theta) = z_{it} (y_{it} - \alpha - \beta_k k_{it} - \beta_l l_{it} - \rho(y_{it-1} - \alpha - \beta_k k_{it-1} - \beta_l l_{it-1})), \quad (\text{B.1})$$

with instrument vector $z_{it} = (l_{it-1}, k_{it}, m_{it-1})$.

C Tables

Table C.1: Monte-Carlo estimates ($\sigma_{\xi_l} = 0.1$)

Meas.	ACF				LP				CUE			
	β_l		β_k		β_l		β_k		β_l		β_k	
	Coef.	S.D.	Coef.	S.D.	Coef.	S.D.	Coef.	S.D.	Coef.	S.D.	Coef.	S.D.
0.000	0.595	0.038	0.405	0.041	0.600	0.010	0.391	0.017	0.599	0.023	0.400	0.007
0.100	0.638	0.068	0.365	0.070	0.907	0.005	2.54e4	2.42e5	0.598	0.031	0.400	0.008
0.200	0.675	0.084	0.329	0.087	0.945	0.004	1.28e6	1.27e6	0.591	0.043	0.402	0.010
0.500	-7.40e8	1.09e10	7.39e8	1.09e10	0.972	0.003	4.41e5	4.00e5	0.581	0.076	0.404	0.015

1000 replications. True values of β_l and β_k are 0.6 and 0.4, respectively. Values in red indicate explosive behavior.

Table C.2: Rejection frequencies at 5 % nominal level ($\sigma_{\xi_l} = 0.10$)

Meas.	Wald	S	DH	CD reject
0.0	0.064	0.060	0.065	1.000
0.1	0.059	0.048	0.079	1.000
0.2	0.075	0.062	0.177	1.000
0.5	0.128	0.050	0.427	0.995

1000 replications. Wald: joint $\chi^2(3)$ at the CUE estimate. S : joint Anderson–Rubin $\chi^2(3)$ evaluated at the true θ_0 . DH: Doornik–Hansen multivariate normality test on the bootstrap distribution. CD reject: rejection frequency of the Cragg–Donald rank test ($T_{CD} \sim \chi^2(1)$), where rejection indicates identification. The design is just identified ($d_\phi = d_\theta = 3$), so the Kleibergen J -statistic has zero degrees of freedom and the KLM statistic coincides with S replication by replication; only S is reported.

Table C.3: Elasticity estimates: Chile ENIA, ISIC 311

Parameter	Estimates		95% CI			Cond. F	BS normality
	ACF	CUE	Wald	Subset- S	Subset- JK	F_j	SW p -value
β_k	0.040	0.014	[0.006, 0.022]	[0.004, 0.025]	[0.006, 0.034]	—	0.698
β_ℓ	0.123	0.110	[0.086, 0.133]	[0.083, 0.143]	[0.058, 0.118]	481	0.556
β_m	0.905	0.948	[0.932, 0.965]	[0.923, 0.968]	[0.934, 0.978]	529	0.303
Returns to scale	1.067	1.072	[1.060, 1.085]	[1.055, 1.091]	[1.030, 1.090]	—	—

Cragg–Donald minimum-eigenvalue statistic: $T_{CD} = 336.1$ (df= 2, $p < 0.001$). Hansen J -test at CUE: $J = 0.258$ (df= 1, $p = 0.612$).

Notes: Wald confidence intervals are based on the cluster-robust variance at the CUE estimate. “Subset- S ” reports identification-robust confidence sets obtained by inverting the S-statistic. “Subset- JK ” reports the combined Kleibergen J–K confidence sets ($\alpha_K = 0.04$, $\alpha_J = 0.01$), which correct for the spurious power decline of the K statistic by pretesting moment validity. Cond. F reports the Sanderson–Windmeijer-style conditional first-stage F_j for the corresponding input quasi-difference (Section 4.2); F_k is omitted because capital enters the instrument set by construction. “SW p -value” reports the bootstrap p -value of a Shapiro–Wilk normality test for the corresponding standardized bootstrap distribution.

Table C.4: Elasticity estimates: U.S. Compustat, NAICS 31

Parameter	Estimates		95% CI			Cond. F	BS normality
	ACF	CUE	Wald	Subset- S	Subset- JK	F_j	SW p -value
β_k	0.329	1.696	[−7.133, 10.524]	[−0.100, 2.500]	[0.200, 2.700]	—	< 0.001
β_ℓ	0.132	−0.992	[−9.601, 7.618]	[−4.790, 1.810]	[<−5.990, 1.110]	0.015	< 0.001
β_m	0.561	−0.537	[−5.200, 4.126]	[−2.540, 0.660]	[<−5.540, 1.260]	0.091	< 0.001
Returns to scale	1.022	0.167	[−4.257, 4.591]	[−0.430, 0.770]	[<−4.830, >5.170]	—	—

Cragg–Donald minimum-eigenvalue statistic: $T_{CD} = 0.093$ (df= 2, $p = 0.955$); fails to reject underidentification. Hansen J -test at CUE: $J = 0.905$ (df= 1, $p = 0.341$).

Notes: Wald confidence intervals are based on the cluster-robust variance at the CUE estimate. “Subset- S ” reports identification-robust confidence sets obtained by inverting the S-statistic. “Subset- JK ” reports the combined Kleibergen J–K confidence sets ($\alpha_K = 0.04$, $\alpha_J = 0.01$). A leading “<” (trailing “>”) indicates that the set extends to the left (right) boundary of the estimation grid; the reported number is the grid boundary used. Cond. F reports the Sanderson–Windmeijer-style conditional first-stage F_j for the corresponding input quasi-difference (Section 4.2); F_k is omitted because capital enters the instrument set by construction. “SW p -value” reports the bootstrap p -value of a Shapiro–Wilk normality test for the corresponding standardized bootstrap distribution.

D Figures

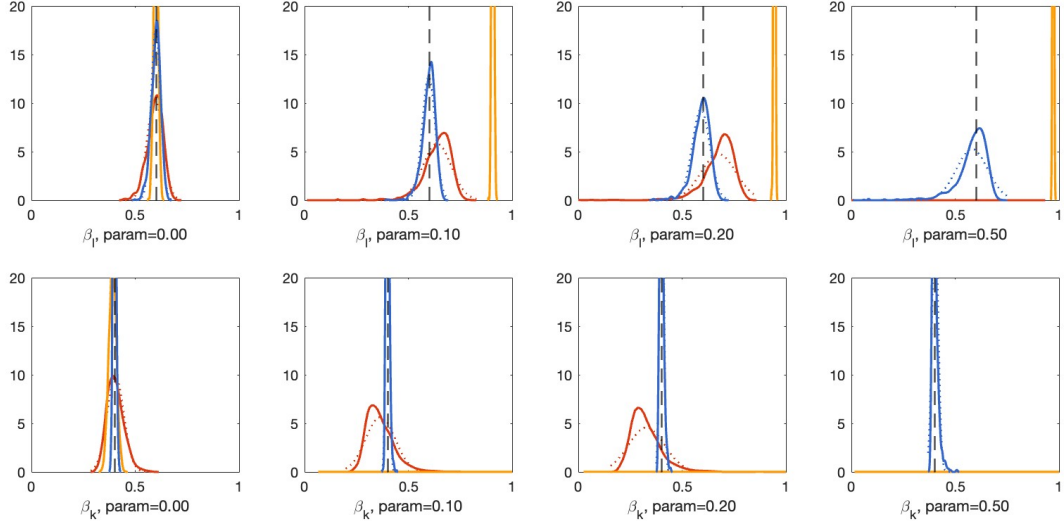


Figure D.1: 1000 replications. $\sigma_{\xi_l} = 0.10$. Empirical densities of the estimates for β_l (top row) and β_k (bottom row) across measurement-error levels 0%, 10%, 20%, 50% (left to right). Methods: ACF (red), LP (orange), CUE (blue). The dashed vertical line marks the true parameter value. Empirical densities: solid. Normal pdf: dotted. 1000 replications.

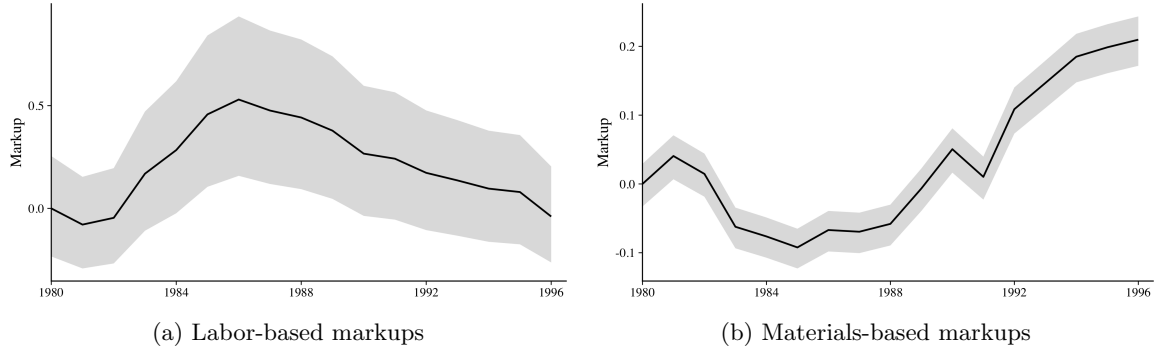


Figure D.2: Chile (ENIA, ISIC 311): changes in mean markups with identification-robust bounds. Solid line: mean plug-in CUE markups. Shaded band: mean bounds induced by identification-robust elasticity intervals. Series are plotted relative to the first sample year.

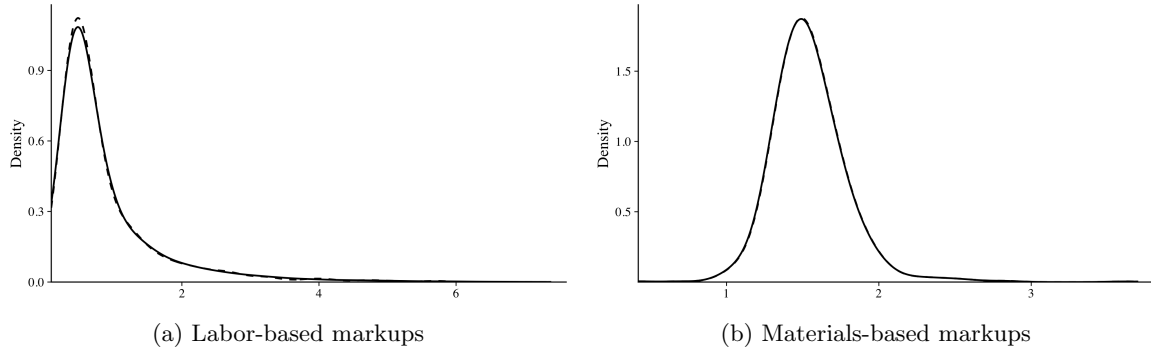


Figure D.3: Chile (ENIA, ISIC 311): markup distributions in 1996. Dashed/red line: imputation-mixture kernel density from draws $\mu_{it}^{v,*} \sim \text{Unif}[\underline{\mu}_{it}^v, \bar{\mu}_{it}^v]$ pooled across repetitions. Solid/blue line: kernel density of plug-in CUE markups $\hat{\mu}_{it}^v$.

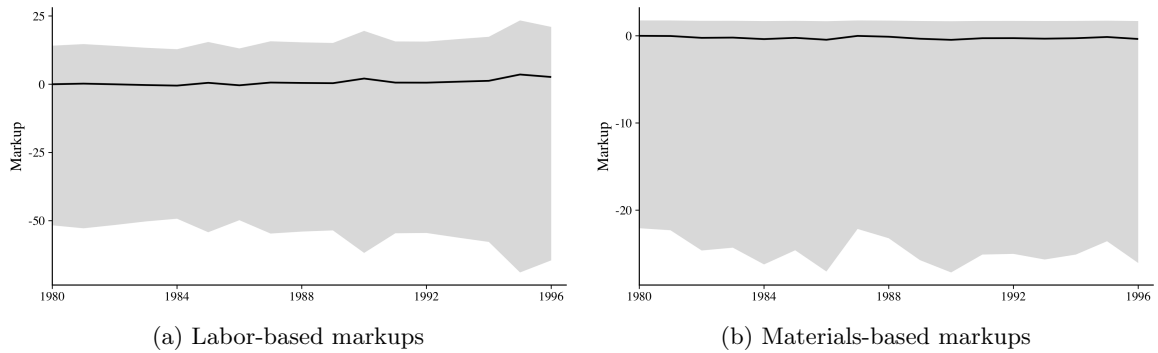


Figure D.4: U.S. (Compustat, NAICS 31): changes in mean markups with identification-robust bounds. Solid line: mean plug-in CUE markups. Shaded band: mean bounds induced by identification-robust elasticity intervals. Series are plotted relative to the first sample year.

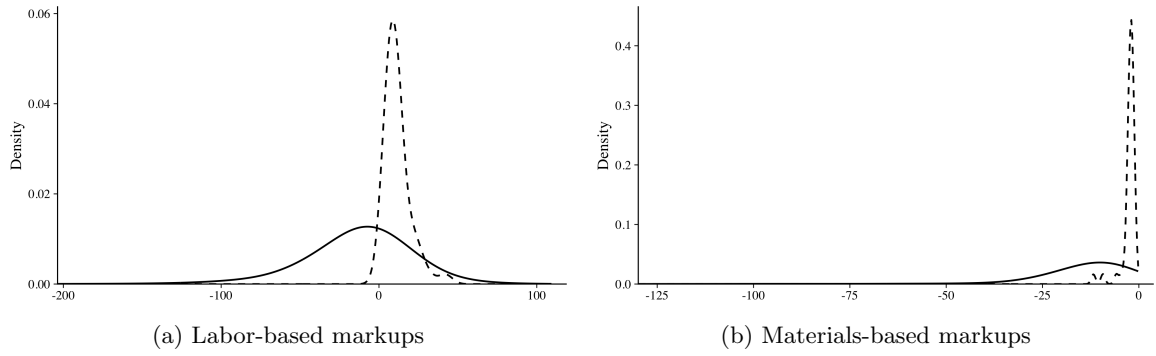


Figure D.5: U.S. (Compustat, NAICS 31): markup distributions in 1996. Dashed line: imputation-mixture kernel density from draws $\mu_{it}^{v,*} \sim \text{Unif}[\underline{\mu}_{it}^v, \overline{\mu}_{it}^v]$ pooled across repetitions. Solid line: kernel density of plug-in CUE markups $\hat{\mu}_{it}^v$.